PYTHON EDA / SALES MARKET

Gurgaon Real Estate Market Analysis

Identifying pricing drivers, market structure, locality intelligence and value opportunities

Source 99acres.com property listings	Focus Residential sales market
Method Python • Pandas • NumPy • Matplotlib • Seaborn	Deliverable Full analytical notebook

Notebook rendered as a LinkedIn-ready document. The analytical content, code, outputs and charts are preserved from the uploaded .ipynb.

Notebook contents

The PDF follows the original notebook sequence so the reader can move from data preparation through validation, exploratory analysis, value-driver analysis and strategic recommendations.

- Gurgaon Real Estate Market Analysis: Identifying Pricing Drivers and Value Opportunities
- Executive Overview
- Business Problem
- Analytical Objectives
- Dataset Overview
- Gurgaon Residential Real Estate Market Analysis: Understanding Pricing Drivers and Identifying Value Opportunities
- Import Libraries
- Data Loading
- Initial Data Exploration
- Feature Selection
- Fixing Datatypes
- Standardizing Price Column
- Initial Missing Value Audit
- Handling Null Values For the Text Columns
- Create Sales Dataset
- Create Rental Dataset
- Feature Engineering
- Data Quality Validation & Anomaly Treatment for Sales Dataset
- Data Quality Validation & Anomaly Treatment
- Exploratory Data Analysis (EDA)
- Pricing Landscape
- Outlier Investigation & Treatment
- Market Structure Analysis
- Locality Intelligence
- Property Value Drivers
- Feature Premium Analysis
- Luxury Market Profiling
- Locality Efficiency Map
- Strategic Recommendations
- Executive Summary

Gurgaon Real Estate Market Analysis: Identifying Pricing Drivers and Value Opportunities

Executive Overview

Gurgaon has emerged as one of India's most dynamic residential real estate markets, characterized by significant variation in property prices, housing configurations, locality performance, and investment potential. For buyers and investors, navigating this complexity often requires more than simply comparing property prices.

This project analyzes over 10,000 residential property listings across 229 Gurgaon localities to uncover the key factors driving property valuations, identify premium and value-oriented micro-markets, and evaluate how location, property characteristics, and market positioning influence housing prices.

The analysis aims to transform raw housing data into actionable market intelligence that can support smarter property-buying and investment decisions.

Business Problem

Real estate decisions are often made using limited information, with buyers focusing primarily on listed prices while overlooking critical factors such as locality performance, price efficiency, property size, and premium market dynamics.

As a result, buyers frequently struggle to answer important questions such as:

- Which localities offer the best value for money?
- What factors have the greatest influence on property prices?
- How do premium markets differ from the broader housing market?
- Which areas justify their pricing through larger homes versus location exclusivity?
- Where do potential investment opportunities exist within Gurgaon?

This project addresses these questions through a structured exploratory analysis of Gurgaon's residential housing market.

Analytical Objectives

The analysis focuses on five key areas:

1. Market Structure Analysis

Understand the composition of Gurgaon's housing market through inventory distribution, property configurations, and pricing characteristics.

2. Pricing Intelligence

Identify pricing patterns, valuation distributions, and the primary factors influencing property prices.

3. Locality Performance Analysis

Compare localities based on inventory concentration, property valuations, and price-per-square-foot performance.

4. Luxury Market Profiling

Investigate the characteristics that distinguish premium residential properties from the broader market.

5. Value Opportunity Identification

Identify localities that offer favorable combinations of property size and pricing, helping buyers evaluate value-for-money opportunities.

Dataset Overview

Source: 99acres Housing Listings

Scope of Analysis

- 10,704 Residential Property Listings
- 229 Gurgaon Localities
- Apartments, Builder Floors, and Independent Houses
- Property Attributes including Area, Price, BHK, Bathrooms, Amenities, Furnishing, and Locality Information

The dataset provides a comprehensive view of Gurgaon's residential housing landscape and enables both market-level and locality-level analysis.

Gurgaon Residential Real Estate Market Analysis: Understanding Pricing Drivers and Identifying Value Opportunities

Executive Overview

Gurgaon is one of India's most active and rapidly evolving residential real estate markets, characterized by significant variation in property prices, housing configurations, locality performance, and investment potential. For buyers and investors, evaluating a property based solely on its listed price often provides an incomplete picture of its true value.

This project analyzes 10,000+ residential property listings across 220+ Gurgaon localities to understand the factors that influence property valuations, identify premium and value-oriented micro-markets, and evaluate how property characteristics and location interact to shape housing prices.

The original dataset contained both property sale listings and rental listings. To ensure analytical consistency and meaningful price comparisons, the dataset was segmented into separate sales and rental datasets. All subsequent analysis in this project focuses exclusively on the residential sales market, allowing for a more accurate evaluation of pricing dynamics, locality performance, and investment opportunities.

Through a combination of data cleaning, missing-value treatment, feature engineering, anomaly validation, and exploratory analysis, the project transforms raw housing listings into actionable market intelligence that can support smarter home-buying and investment decisions.

Business Problem

Residential property decisions are often made using limited information, with buyers focusing primarily on total property price while overlooking important factors such as locality performance, price-per-square-foot efficiency, property size, configuration, and market positioning.

As a result, buyers frequently struggle to answer critical questions such as:

- Which factors have the greatest influence on property prices?
- Which localities command genuine pricing power?
- Where can buyers find better space-to-price value?

- How do luxury markets differ from the broader housing market?
- Which localities offer the strongest balance between affordability, space, and long-term potential?

This project addresses these questions through a structured analysis of Gurgaon's residential property sales market.

Analytical Objectives

The analysis focuses on five key areas:

1. Market Structure Analysis

Understand the composition of Gurgaon's residential housing market through inventory distribution, pricing patterns, and property configurations.

2. Pricing Driver Analysis

Identify the property attributes that most strongly influence housing prices and evaluate how area, configuration, and other characteristics contribute to valuation.

3. Locality Performance Analysis

Compare Gurgaon localities based on inventory concentration, pricing levels, and price-per-square-foot performance.

4. Luxury Market Profiling

Investigate the characteristics that distinguish premium residential properties from the broader housing market.

5. Value Opportunity Identification

Identify localities that provide favorable combinations of living space, pricing efficiency, and investment potential.

Dataset Overview

Source: 99acres Residential Property Listings

Scope of Analysis

- 10,000+ Residential Property Listings
- 220+ Gurgaon Localities
- Sale and Rental Listings Available in the Original Dataset
- Apartments, Builder Floors, and Independent Houses
- Property Attributes Including Area, Price, BHK Configuration, Bathrooms, Amenities, Furnishing, Property Type, and Locality Information

Data Preparation

The original dataset was first segmented into separate sales and rental datasets. Since sale and rental properties follow fundamentally different pricing dynamics, combining them would distort valuation metrics, pricing distributions, and market insights. Therefore, all exploratory analysis was conducted exclusively on the sales dataset.

The data then underwent a structured preparation process that included:

- Missing-value treatment
- Feature engineering
- Data-quality validation
- Area-based anomaly detection

- Price-per-square-foot anomaly treatment

Properties with unrealistic area values and extreme lower-tail price-per-square-foot observations were identified and removed to improve analytical reliability and ensure that the final dataset accurately reflected Gurgaon’s residential sales market.

The resulting analytical dataset provides a robust foundation for understanding pricing behavior, locality performance, luxury market dynamics, and value-oriented investment opportunities within Gurgaon’s residential housing market.

Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

pd.set_option('display.max_columns', None)
```

Data Loading

```
data_frame= pd.read_csv(r"D:\DATA analysis\projects\Key Projects\99 acre\Gurgaon Data\gurgaon.csv")
df=data_frame.copy()
df.head()
```

```
     PROP_ID                                          PHOTO_URL  \
0  O71585872  https://mediacdn.99acres.com/media1/22409/13/4...
1  T71585466  https://mediacdn.99acres.com/media1/22425/6/44...
2  D71549524  https://mediacdn.99acres.com/media1/22386/13/4...
3  I71493958  https://mediacdn.99acres.com/media1/22353/14/4...
4  H71479412  https://mediacdn.99acres.com/media1/22352/4/44...

                                MEDIUM_PHOTO_URL PREFERENCE  \
0  https://mediacdn.99acres.com/media1/22409/13/4...          S
1  https://mediacdn.99acres.com/media1/22425/6/44...          S
2  https://mediacdn.99acres.com/media1/22386/13/4...          R
3  https://mediacdn.99acres.com/media1/22353/14/4...          S
4  https://mediacdn.99acres.com/media1/22352/4/44...          R

                                     DESCRIPTION          PROPERTY_TYPE  \
0   4 bhk flat for resale in alpha corp gurgaon one  Residential Apartment
1  Explore dlf ultima, an upscale ready-To-Move h...  Residential Apartment
2  A 3 bhk flat for rent in sector 66 gurgaon clo...  Residential Apartment
3  Looking for a 3 bhk property for purchase in g...  Residential Apartment
4  This 4 bhk flat in sector 104 gurgaon is avail...  Residential Apartment

      CITY            LOCALITY  TRANSACT_TYPE  OWNTYPE  BEDROOM_NUM  \
0  Gurgaon   Sector 84 Gurgaon            1.0        1          4.0
1  Gurgaon   Sector 81 Gurgaon            1.0        1          4.0
2  Gurgaon   Sector 66 Gurgaon            NaN        0          3.0
3  Gurgaon  Sector 112 Gurgaon            1.0        1          3.0
4  Gurgaon  Sector 104 Gurgaon            NaN        0          3.0

   BATHROOM_NUM  BALCONY_NUM  PRICE_PER_UNIT_AREA  FURNISH  FACING  AGE  \
0             4          4.0               8766.0        4       1    2
1             4          3.0              21176.0        4       1    1
2             3          2.0                 43.0        1       4    1
3             3          3.0              13740.0        4       0    1
4             4          4.0                 17.0        4       6    1

  FLOOR_NUM  TOTAL_FLOOR                                       FEATURES  \
0        14         26.0  33,23,12,46,25,47,39,1,3,6,9,40,41,30,20,42,21,32
1         7         29.0  44,33,23,12,46,25,47,26,17,29,19,1,3,5,6,9,30,...
2         7         17.0  44,23,45,24,46,25,47,28,29,30,31,32,33,12,17,3...
3        14         28.0       33,23,12,25,26,1,2,3,5,6,9,40,30,41,31,20,21
4        12         27.0   44,23,12,46,25,26,28,1,2,3,6,9,40,30,41,31,20,21

    REGISTER_DATE                PROP_NAME  MIN_PRICE  MAX_PRICE  PRICE_SQFT  \
0  29th Sep, 2023  Alpha Corp GurgaonOne 84   26300000   26300000      8766.0
1  29th Sep, 2023            DLF The Ultima   36000000   36000000     21176.0
2  27th Sep, 2023  Emaar MGF The Palm Drive      85000      85000        43.0
3  23rd Sep, 2023       Experion Windchants   38500000   38500000     13740.0
4  23rd Sep, 2023               ATS Triumph      40000      40000        17.0

  LISTING  BUILDING_ID  CARPET_SQFT  SUPERBUILTUP_SQFT  BROKERAGE  \
0       R        10532       3000.0             3434.0   263000.0
1       R        12704       1700.0             2870.0        1.0
```

```
2       R         3252       1450.0             1950.0    42000.0
3       R         9772          NaN             2802.0        1.0
4       R        41176          NaN             2290.0        1.0

                                       MAP_DETAILS            FSL_Data  \
0  {'LATITUDE': '28.4065341', 'LONGITUDE': '76.96...  {'LOCALITY_ID': 0}
1  {'LATITUDE': '28.3889069', 'LONGITUDE': '76.95...  {'LOCALITY_ID': 0}
2  {'LATITUDE': '28.401406', 'LONGITUDE': '77.050...  {'LOCALITY_ID': 0}
3  {'LATITUDE': '28.518677', 'LONGITUDE': '77.015...  {'LOCALITY_ID': 0}
4  {'LATITUDE': '28.47844', 'LONGITUDE': '76.9861...  {'LOCALITY_ID': 0}

   MIN_AREA_SQFT  MAX_AREA_SQFT  \
0     319.028902     319.028902
1     266.631610     266.631610
2     181.160850     181.160850
3     260.314206     260.314206
4     212.747870     212.747870

                                        FORMATTED  \
0  {'AVAIL': {'AVAILABILITY': 2, 'SUB_AVAILABILIT...
1  {'AVAIL': {'AVAILABILITY': 2, 'SUB_AVAILABILIT...
2  {'AVAIL': {'AVAILABILITY': 2, 'SUB_AVAILABILIT...
3  {'AVAIL': {'AVAILABILITY': 2, 'SUB_AVAILABILIT...
4  {'AVAIL': {'AVAILABILITY': 2, 'SUB_AVAILABILIT...

                                        AMENITIES  \
0  20,21,32,23,46,47,39,1,6,9,40,41,30,42,33,12,2...
1  17,5,20,21,10,44,23,46,47,29,19,1,6,9,30,41,43...
2  32,17,2,5,20,21,44,23,45,24,46,47,28,29,30,39,...
3  2,5,20,21,23,1,6,9,40,30,41,33,12,25,26,3,31,1...
4   2,20,21,44,23,46,28,1,6,9,40,30,41,12,25,26,3,31

                                         TOP_USPS  \
0  ['North Facing', 'Recently Renovated', 'Gated ...
1  ['North Facing', "st. xavier's within 500m", '...
2  ['3 Baths', 'Full Power Backup', 'huda city ce...
3  ['Gated Society', 'Overlooking Park/Garden', '...
4  ['4 Baths', 'Full Power Backup', 'Gated Societ...

                                           PD_URL          EXPIRY_DATE  \
0  /4-bhk-bedroom-apartment-flat-for-sale-in-alph...  2023-11-28 00:00:00
1  /4-bhk-bedroom-apartment-flat-for-sale-in-dlf-...  2023-11-28 00:00:00
2  /3-bhk-bedroom-apartment-flat-for-rent-in-emaa...  2023-11-26 00:00:00
3  /3-bhk-bedroom-apartment-flat-for-sale-in-expe...  2023-11-22 00:00:00
4  /3-bhk-bedroom-apartment-flat-for-rent-in-ats-...  2023-11-21 00:00:00

                GROUP_NAME         AREA    PRICE  \
0  SPID_GROUP_SEARCH_NORMAL  3434 sq.ft.  2.63 Cr
1  SPID_GROUP_SEARCH_NORMAL  2870 sq.ft.   3.6 Cr
2  SPID_GROUP_SEARCH_NORMAL  1950 sq.ft.   85,000
3  SPID_GROUP_SEARCH_NORMAL  2802 sq.ft.  3.85 Cr
4  SPID_GROUP_SEARCH_NORMAL  2290 sq.ft.   40,000

                      PROP_HEADING  \
0   4 BHK Flat in Sector 84 Gurgaon
1   4 BHK Flat in Sector 81 Gurgaon
2   3 BHK Flat in Sector 66 Gurgaon
3  3 BHK Flat in Sector 112 Gurgaon
4  3 BHK Flat in Sector 104 Gurgaon

                                  PROP_DETAILS_URL  \
0  4-bhk-bedroom-apartment-flat-for-sale-in-alpha...
1  4-bhk-bedroom-apartment-flat-for-sale-in-dlf-t...
2  3-bhk-bedroom-apartment-flat-for-rent-in-emaar...
3  3-bhk-bedroom-apartment-flat-for-sale-in-exper...
4  3-bhk-bedroom-apartment-flat-for-rent-in-ats-t...

               CLASS_HEADING CLASS_LABEL  \
0       Dealer: propertyline      Dealer
1  Dealer: MVD AND ASSOCIATES      Dealer
2   Dealer: 1st Nest Realtors      Dealer
3          Dealer: GROW MORE      Dealer
4          Dealer: GROW MORE      Dealer

                       SECONDARY_TAGS  \
0         ['READY TO MOVE', 'RESALE']
1  ['READY TO MOVE', 'RESALE', 'RERA']
2                       ['FURNISHED']
3  ['READY TO MOVE', 'RESALE', 'RERA']
4                                  []

                                  PROPERTY_IMAGES  \
0  ['https://mediacdn.99acres.com/media1/22409/13...
1  ['https://mediacdn.99acres.com/media1/22425/6/...
2  ['https://mediacdn.99acres.com/media1/22386/13...
3  ['https://mediacdn.99acres.com/media1/22353/14...
```

```
4  ['https://mediacdn.99acres.com/media1/22352/4/...

                                THUMBNAIL_IMAGES  TOTAL_LANDMARK_COUNT  \
0  ['https://mediacdn.99acres.com/media1/22409/13...                  10.0
1  ['https://mediacdn.99acres.com/media1/22425/6/...                   9.0
2  ['https://mediacdn.99acres.com/media1/22386/13...                  14.0
3  ['https://mediacdn.99acres.com/media1/22353/14...                   8.0
4  ['https://mediacdn.99acres.com/media1/22352/4/...                  14.0

                        FORMATTED_LANDMARK_DETAILS   CONTACT_NAME  \
0  [{'category': 'Shopping', 'text': '4 Shoppings...  gaurav khosla
1  [{'category': 'MetroStation', 'text': '1 Metro...          Kamal
2  [{'category': 'MetroStation', 'text': '1 Metro...        Sandeep
3  [{'category': 'Shopping', 'text': '1 Shopping'...          Kohli
4  [{'category': 'MetroStation', 'text': '1 Metro...          Kohli

  CONTACT_COMPANY_NAME                                DEALER_PHOTO_URL  \
0         propertyline  https://static.99acres.com/images/dealer_pnava...
1   MVD AND ASSOCIATES  https://static.99acres.com/images/dealer_pnava...
2    1st Nest Realtors  https://static.99acres.com/images/dealer_pnava...
3            GROW MORE  https://mediacdn.99acres.com/media1/17762/9/35...
4            GROW MORE  https://mediacdn.99acres.com/media1/17762/9/35...

              SOCIETY_NAME             BUILDING_NAME  CITY_ID  \
0  Alpha Corp GurgaonOne 84  Alpha Corp GurgaonOne 84        8
1            DLF The Ultima            DLF The Ultima        8
2  Emaar MGF The Palm Drive  Emaar MGF The Palm Drive        8
3       Experion Windchants       Experion Windchants        8
4               ATS Triumph               ATS Triumph        8

  LOCALITY_WO_CITY                                          profile  \
0        Sector 84  {'CONTACT_NAME': 'gaurav khosla', 'CONTACT_COM...
1        Sector 81  {'CONTACT_NAME': 'Kamal', 'CONTACT_COMPANY_NAM...
2        Sector 66  {'CONTACT_NAME': 'Sandeep', 'CONTACT_COMPANY_N...
3       Sector 112  {'CONTACT_NAME': 'Kohli', 'CONTACT_COMPANY_NAM...
4       Sector 104  {'CONTACT_NAME': 'Kohli', 'CONTACT_COMPANY_NAM...

                                              xid  \
0  {'PROJECTID': 5096, 'PROJ_NAME': 'Alpha Corp G...
1  {'PROJECTID': 7182, 'PROJ_NAME': 'DLF The Ulti...
2  {'PROJECTID': 19, 'PROJ_NAME': 'Emaar MGF The ...
3  {'PROJECTID': 4342, 'PROJ_NAME': 'Experion Win...
4  {'PROJECTID': 25860, 'PROJ_NAME': 'ATS Triumph...

                                          metadata  \
0  {'PROP_PHOTO_COUNT': '4', 'PROP_VIDEO_COUNT': ...
1  {'PROP_PHOTO_COUNT': '4', 'PROP_VIDEO_COUNT': ...
2  {'PROP_PHOTO_COUNT': '1', 'PROP_VIDEO_COUNT': ...
3  {'PROP_PHOTO_COUNT': '1', 'PROP_VIDEO_COUNT': ...
4  {'PROP_PHOTO_COUNT': '14', 'PROP_VIDEO_COUNT':...

                                          location  BUILTUP_SQFT  \
0  {'CITY': '8', 'CITY_NAME': 'Gurgaon', 'BUILDIN...           NaN
1  {'CITY': '8', 'CITY_NAME': 'Gurgaon', 'BUILDIN...           NaN
2  {'CITY': '8', 'CITY_NAME': 'Gurgaon', 'BUILDIN...       1920.00
3  {'CITY': '8', 'CITY_NAME': 'Gurgaon', 'BUILDIN...       2225.56
4  {'CITY': '8', 'CITY_NAME': 'Gurgaon', 'BUILDIN...           NaN

   SUPER_SQFT  COMMON_FURNISHING_ATTRIBUTES  QUALITY_SCORE  \
0         NaN                           NaN            NaN
1         NaN                           NaN            NaN
2         NaN                           NaN            NaN
3         NaN                           NaN            NaN
4         NaN                           NaN            NaN

  FURNISHING_ATTRIBUTES
0                   NaN
1                   NaN
2                   NaN
3                   NaN
4                   NaN
```

Initial Data Exploration

```
df.shape
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10704 entries, 0 to 10703
Data columns (total 67 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   PROP_ID                       10704 non-null  object
 1   PHOTO_URL                     8864 non-null   object
```

```
 2   MEDIUM_PHOTO_URL              8864 non-null   object
 3   PREFERENCE                    10704 non-null  object
 4   DESCRIPTION                   10704 non-null  object
 5   PROPERTY_TYPE                 10704 non-null  object
 6   CITY                          10704 non-null  object
 7   LOCALITY                      10704 non-null  object
 8   TRANSACT_TYPE                 10152 non-null  float64
 9   OWNTYPE                       10704 non-null  int64
 10  BEDROOM_NUM                   10511 non-null  float64
 11  BATHROOM_NUM                  10704 non-null  int64
 12  BALCONY_NUM                   10501 non-null  float64
 13  PRICE_PER_UNIT_AREA           10704 non-null  float64
 14  FURNISH                       10704 non-null  int64
 15  FACING                        10704 non-null  int64
 16  AGE                           10704 non-null  int64
 17  FLOOR_NUM                     9804 non-null   object
 18  TOTAL_FLOOR                   10697 non-null  float64
 19  FEATURES                      10704 non-null  object
 20  REGISTER_DATE                 10704 non-null  object
 21  PROP_NAME                     9135 non-null   object
 22  MIN_PRICE                     10704 non-null  int64
 23  MAX_PRICE                     10704 non-null  int64
 24  PRICE_SQFT                    10704 non-null  float64
 25  LISTING                       10704 non-null  object
 26  BUILDING_ID                   10704 non-null  int64
 27  CARPET_SQFT                   6539 non-null   float64
 28  SUPERBUILTUP_SQFT             6712 non-null   float64
 29  BROKERAGE                     10623 non-null  float64
 30  MAP_DETAILS                   10704 non-null  object
 31  FSL_Data                      10704 non-null  object
 32  MIN_AREA_SQFT                 10704 non-null  float64
 33  MAX_AREA_SQFT                 10704 non-null  float64
 34  FORMATTED                     10704 non-null  object
 35  AMENITIES                     10620 non-null  object
 36  TOP_USPS                      8179 non-null   object
 37  PD_URL                        10702 non-null  object
 38  EXPIRY_DATE                   10704 non-null  object
 39  GROUP_NAME                    10704 non-null  object
 40  AREA                          10704 non-null  object
 41  PRICE                         10704 non-null  object
 42  PROP_HEADING                  10704 non-null  object
 43  PROP_DETAILS_URL              10704 non-null  object
 44  CLASS_HEADING                 10704 non-null  object
 45  CLASS_LABEL                   10704 non-null  object
 46  SECONDARY_TAGS                10704 non-null  object
 47  PROPERTY_IMAGES               8863 non-null   object
 48  THUMBNAIL_IMAGES              8863 non-null   object
 49  TOTAL_LANDMARK_COUNT          10638 non-null  float64
 50  FORMATTED_LANDMARK_DETAILS    10638 non-null  object
 51  CONTACT_NAME                  10660 non-null  object
 52  CONTACT_COMPANY_NAME          10699 non-null  object
 53  DEALER_PHOTO_URL              10704 non-null  object
 54  SOCIETY_NAME                  9135 non-null   object
 55  BUILDING_NAME                 9135 non-null   object
 56  CITY_ID                       10704 non-null  int64
 57  LOCALITY_WO_CITY              10704 non-null  object
 58  profile                       10704 non-null  object
 59  xid                           10704 non-null  object
 60  metadata                      10704 non-null  object
 61  location                      10704 non-null  object
 62  BUILTUP_SQFT                  2733 non-null   float64
 63  SUPER_SQFT                    875 non-null    float64
 64  COMMON_FURNISHING_ATTRIBUTES  0 non-null      float64
 65  QUALITY_SCORE                 2 non-null      float64
 66  FURNISHING_ATTRIBUTES         2 non-null      object
dtypes: float64(16), int64(9), object(42)
memory usage: 5.5+ MB
```

	TRANSACT_TYPE	OWNTYPE	BEDROOM_NUM	BATHROOM_NUM	BALCONY_NUM
count	10152.000000	10704.000000	10511.000000	10704.000000	10501.000000
mean	1.001182	0.968610	3.327181	3.484118	2.930959
std	0.034362	0.304357	1.288234	1.518250	0.916045
min	1.000000	0.000000	1.000000	0.000000	0.000000
25%	1.000000	1.000000	3.000000	3.000000	2.000000
50%	1.000000	1.000000	3.000000	3.000000	3.000000
75%	1.000000	1.000000	4.000000	4.000000	4.000000
max	2.000000	4.000000	22.000000	22.000000	4.000000

	PRICE_PER_UNIT_AREA	FURNISH	FACING	AGE
count	1.070400e+04	10704.000000	10704.000000	10704.000000
mean	4.919281e+04	3.298954	3.460762	3.228886
std	9.227134e+05	1.231244	2.049320	2.154224
min	0.000000e+00	0.000000	0.000000	0.000000
25%	8.371500e+03	2.000000	1.000000	1.000000
50%	1.191750e+04	4.000000	3.000000	2.000000
75%	1.771950e+04	4.000000	5.000000	6.000000

```
max           8.500000e+07      4.000000      8.000000      6.000000

        TOTAL_FLOOR     MIN_PRICE     MAX_PRICE    PRICE_SQFT   BUILDING_ID  \
count  10697.000000  1.070400e+04  1.070400e+04  1.070400e+04  1.070400e+04
mean      14.787137  3.184154e+07  3.184388e+07  4.919281e+04  2.284910e+05
std       11.353743  5.168846e+07  5.168949e+07  9.227134e+05  3.789679e+05
min        0.000000  0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00
25%        4.000000  1.389275e+07  1.389275e+07  8.371500e+03  6.630000e+02
50%       14.000000  2.200000e+07  2.200000e+07  1.191750e+04  1.192800e+04
75%       22.000000  3.700000e+07  3.700000e+07  1.771950e+04  3.407790e+05
max       51.000000  2.147484e+09  2.147484e+09  8.500000e+07  1.144346e+06

         CARPET_SQFT  SUPERBUILTUP_SQFT     BROKERAGE  MIN_AREA_SQFT  \
count    6539.000000        6712.000000  1.062300e+04   10704.000000
mean     2483.509454        2361.503528  4.915147e+04     330.136721
std     16707.402451        3382.475458  7.898489e+05    7191.202936
min         0.000000           0.000000  0.000000e+00       2.787090
25%      1200.000000        1620.000000  1.000000e+00     145.857710
50%      1650.000000        2050.000000  1.000000e+00     192.495016
75%      2350.000000        2690.000000  1.000000e+00     252.603257
max    577946.000000      258879.000000  7.310000e+07  728433.842400

       MAX_AREA_SQFT  TOTAL_LANDMARK_COUNT  CITY_ID  BUILTUP_SQFT  \
count   10704.000000          10638.000000  10704.0   2733.000000
mean      330.142935             20.717992      8.0   2334.236483
std      7191.202775             15.659258      0.0   2879.712010
min         2.787090              1.000000      8.0      0.000000
25%       145.857710             10.000000      8.0   1500.000000
50%       192.495016             13.000000      8.0   1900.000000
75%       252.603257             44.000000      8.0   2599.000000
max    728433.842400             50.000000      8.0  67500.000000

         SUPER_SQFT  COMMON_FURNISHING_ATTRIBUTES  QUALITY_SCORE
count  8.750000e+02                           0.0            2.0
mean   1.318212e+04                           NaN           95.0
std    2.666606e+05                           NaN            0.0
min    1.000000e+02                           NaN           95.0
25%    1.512000e+03                           NaN           95.0
50%    2.403000e+03                           NaN           95.0
75%    3.240000e+03                           NaN           95.0
max    7.840800e+06                           NaN           95.0
```

Feature Selection

The original dataset contains 67 columns covering various aspects of property listings. However, not all variables are required for the objectives of this analysis.

To improve analytical efficiency, reduce noise, and focus on the most business-relevant information, a subset of key features is selected. The retained columns capture the essential dimensions of the property market, including:

- Property identification
- Location information
- Property characteristics
- Size and pricing metrics
- Furnishing and property condition
- Property descriptions and amenities
- Society/building information

By narrowing the dataset to these core variables, we create a cleaner and more focused analytical dataset that is easier to explore, visualize, and derive meaningful insights from.

```
selected_cols = [
    # Core identifiers
    'PROP_ID','PROP_NAME','PREFERENCE' ,

    # Location
    'CITY', 'LOCALITY',

    # Property details
    'PROPERTY_TYPE', 'BEDROOM_NUM', 'BATHROOM_NUM',
    'BALCONY_NUM', 'FLOOR_NUM', 'TOTAL_FLOOR',

    # Size & pricing
    'AREA', 'PRICE',

    # Additional features
```

```
    'FURNISH', 'FACING', 'AGE','BROKERAGE', 'CONTACT_COMPANY_NAME','CLASS_LABEL',

    # Society / building
    'SOCIETY_NAME'
]

df = df[selected_cols]

print(df.head())
```

```
     PROP_ID                PROP_NAME PREFERENCE     CITY  \
0  O71585872  Alpha Corp GurgaonOne 84          S  Gurgaon
1  T71585466            DLF The Ultima          S  Gurgaon
2  D71549524  Emaar MGF The Palm Drive          R  Gurgaon
3  I71493958       Experion Windchants          S  Gurgaon
4  H71479412               ATS Triumph          R  Gurgaon

            LOCALITY          PROPERTY_TYPE  BEDROOM_NUM  BATHROOM_NUM  \
0   Sector 84 Gurgaon  Residential Apartment          4.0             4
1   Sector 81 Gurgaon  Residential Apartment          4.0             4
2   Sector 66 Gurgaon  Residential Apartment          3.0             3
3  Sector 112 Gurgaon  Residential Apartment          3.0             3
4  Sector 104 Gurgaon  Residential Apartment          3.0             4

   BALCONY_NUM FLOOR_NUM  TOTAL_FLOOR         AREA    PRICE  FURNISH  FACING  \
0          4.0        14         26.0  3434 sq.ft.  2.63 Cr        4       1
1          3.0         7         29.0  2870 sq.ft.   3.6 Cr        4       1
2          2.0         7         17.0  1950 sq.ft.   85,000        1       4
3          3.0        14         28.0  2802 sq.ft.  3.85 Cr        4       0
4          4.0        12         27.0  2290 sq.ft.   40,000        4       6

   AGE  BROKERAGE CONTACT_COMPANY_NAME CLASS_LABEL              SOCIETY_NAME
0    2   263000.0         propertyline      Dealer  Alpha Corp GurgaonOne 84
1    1        1.0   MVD AND ASSOCIATES      Dealer            DLF The Ultima
2    1    42000.0    1st Nest Realtors      Dealer  Emaar MGF The Palm Drive
3    1        1.0            GROW MORE      Dealer       Experion Windchants
4    1        1.0            GROW MORE      Dealer               ATS Triumph
```

Dropped Columns:

- Image URLs → not useful for analysis
- Duplicate area columns → redundancy
- Metadata / JSON fields → complex, not needed
- Contact info → irrelevant

Fixing Datatypes

```
df.dtypes
```

```
PROP_ID                 object
PROP_NAME               object
PREFERENCE              object
CITY                    object
LOCALITY                object
PROPERTY_TYPE           object
BEDROOM_NUM            float64
BATHROOM_NUM             int64
BALCONY_NUM            float64
FLOOR_NUM               object
TOTAL_FLOOR            float64
AREA                    object
PRICE                   object
FURNISH                  int64
FACING                   int64
AGE                      int64
BROKERAGE              float64
CONTACT_COMPANY_NAME    object
CLASS_LABEL             object
SOCIETY_NAME            object
dtype: object
```

```
df['FLOOR_NUM'] = pd.to_numeric(df['FLOOR_NUM'], errors='coerce')
```

Standardizing Area Column

```
df['AREA'] = df['AREA'].str.replace('sq.ft.', '', regex=False)
df['AREA'] = df['AREA'].str.replace(',', '')
df['AREA'] = pd.to_numeric(df['AREA'], errors='coerce')
```

Standardizing Price Column

```
def clean_price(x):
    x = str(x).lower().replace(',', '').replace('₹', '').strip()

    if x in ['price on request', 'call for price', '', 'nan', '—']:
        return None

    try:
        if 'cr' in x:
            return float(x.replace('cr', '').strip()) * 10000000
        elif 'lac' in x:
            return float(x.replace('lac', '').strip()) * 100000
        else:
            return float(x)
    except:
        return None

df['PRICE'] = df['PRICE'].apply(clean_price)
```

Data Cleaning

The PRICE and AREA columns contained mixed units and text values (e.g., "2.63 Cr", "3434 sq.ft.").

To ensure accurate analysis:

- Extracted numeric values
- Standardized price into INR
- Removed unit text from area

This step was critical to avoid incorrect aggregations.

Initial Missing Value Audit

```
df.isnull().sum().sort_values(ascending=False)
```

```
PROP_NAME               1569
SOCIETY_NAME            1569
FLOOR_NUM               1326
PRICE                   1186
BALCONY_NUM              203
BEDROOM_NUM              193
BROKERAGE                 81
TOTAL_FLOOR                7
CONTACT_COMPANY_NAME       5
AREA                       5
PROP_ID                    0
PREFERENCE                 0
CITY                       0
PROPERTY_TYPE              0
BATHROOM_NUM               0
LOCALITY                   0
AGE                        0
FACING                     0
FURNISH                    0
CLASS_LABEL                0
dtype: int64
```

Handling Null Values For the Text Columns

PROP_NAME (Nulls - 1569)

Replaced nulls by 'unknown'

```
df['PROP_NAME'] = df['PROP_NAME'].fillna('Unknown')
```

SOCIETY_NAME (Nulls - 1569)

Replaced nulls by 'unknown'

```
df['SOCIETY_NAME'] = df['SOCIETY_NAME'].fillna('Unknown')
```

CONTACT_COMPANY_NAME (Nulls - 5)

Replaced nulls by 'unknown'

```
df['CONTACT_COMPANY_NAME'] = df['CONTACT_COMPANY_NAME'].fillna('Unknown')

df.isnull().sum().sort_values(ascending=False)
```

```
FLOOR_NUM               1326
PRICE                   1186
BALCONY_NUM              203
BEDROOM_NUM              193
BROKERAGE                 81
TOTAL_FLOOR                7
AREA                       5
PROP_ID                    0
CITY                       0
PREFERENCE                 0
PROP_NAME                  0
BATHROOM_NUM               0
PROPERTY_TYPE              0
LOCALITY                   0
FACING                     0
FURNISH                    0
AGE                        0
CONTACT_COMPANY_NAME       0
CLASS_LABEL                0
SOCIETY_NAME               0
dtype: int64
```

Create Sales Dataset

```
df_sales = df[df['PREFERENCE'] == 'S'].copy()
```

Create Rental Dataset

```
df_rent = df[df['PREFERENCE'] == 'R'].copy()

print("Original Dataset :", len(df))
print("Sales Listings   :", len(df_sales))
print("Rental Listings  :", len(df_rent))
print("Check Total      :", len(df_sales) + len(df_rent))
```

```
Original Dataset : 10704
Sales Listings   : 10155
Rental Listings  : 549
Check Total      : 10704
```

Market Segmentation: Sales vs Rental Listings

The dataset contains both property sale and rental listings. Since pricing behavior, valuation metrics, and buyer objectives differ significantly between these two markets, combining them can lead to misleading insights.

To ensure analytical consistency, the dataset was segmented into separate sales and rental datasets.

SALES DATASET - MISSING VALUE TREATMENT

```
df_sales.isnull().sum()[df_sales.isnull().sum() > 0]
```

```
BEDROOM_NUM     191
BALCONY_NUM     201
FLOOR_NUM      1269
TOTAL_FLOOR       7
AREA              5
PRICE          1077
BROKERAGE        72
dtype: int64
```

```
# PRICE
df_sales['PRICE'] = (
    df_sales['PRICE']
    .fillna(df_sales['PRICE'].median())
)


# AREA
df_sales = df_sales.dropna(subset=['AREA'])

# Bedroom Count
df_sales['BEDROOM_NUM'] = (
    df_sales['BEDROOM_NUM']
    .fillna(df_sales['BEDROOM_NUM'].mode()[0])
)

# Balcony Count
df_sales['BALCONY_NUM'] = (
    df_sales['BALCONY_NUM']
    .fillna(0)
)
```

```
# Floor Number
df_sales['FLOOR_NUM'] = (
    df_sales['FLOOR_NUM']
    .fillna(df_sales['FLOOR_NUM'].median())
)

# Total Floors
df_sales['TOTAL_FLOOR'] = (
    df_sales['TOTAL_FLOOR']
    .fillna(df_sales['TOTAL_FLOOR'].median())
)

# Brokerage
df_sales['BROKERAGE'] = (
    df_sales['BROKERAGE']
    .fillna(0)
)

# Validation
print("Remaining Missing Values:")
print(df_sales.isnull().sum()[df_sales.isnull().sum() > 0])
```

```
Remaining Missing Values:
Series([], dtype: int64)
```

Missing Value Treatment (Sales Dataset)

Missing values were handled using business-aware imputation techniques to maximize data retention while preserving the integrity of the sales market analysis.

- Price values were imputed using the median because price is a highly skewed variable. Median imputation minimizes the influence of extreme luxury properties while preventing the loss of a significant number of records.
- Area is a critical property attribute used in valuation and price-per-square-foot calculations. Since only a very small number of records had missing area values, these records were removed rather than imputed.
- Bedroom Count was imputed using the mode because bedroom configuration is a discrete categorical-like property attribute, and the most frequently occurring value provides the most realistic replacement.
- Balcony Count was replaced with 0, assuming that missing balcony information either indicates the absence of a balcony or that the feature was not specified in the listing.
- Floor Number and Total Floors were imputed using the median to reduce the impact of outliers while preserving the overall distribution of vertical housing characteristics.
- Brokerage values were replaced with 0, treating missing values as unavailable or non-applicable brokerage information.

This approach balances data quality and record retention, ensuring that the sales dataset remains suitable for subsequent feature engineering, valuation analysis, and exploratory data analysis.

RENTAL DATASET - MISSING VALUE TREATMENT

```
df_rent.isnull().sum()[df_rent.isnull().sum() > 0]
```

```
BEDROOM_NUM      2
BALCONY_NUM      2
FLOOR_NUM       57
PRICE          109
BROKERAGE        9
dtype: int64
```

```
# PRICE
df_rent['PRICE'] = (
    df_rent['PRICE']
    .fillna(df_rent['PRICE'].median())
)

# BEDROOM_NUM
df_rent['BEDROOM_NUM'] = (
    df_rent['BEDROOM_NUM']
    .fillna(df_rent['BEDROOM_NUM'].mode()[0])
)

# BALCONY_NUM
df_rent['BALCONY_NUM'] = (
    df_rent['BALCONY_NUM']
```

```
    .fillna(0)
)

# FLOOR_NUM
df_rent['FLOOR_NUM'] = (
    df_rent['FLOOR_NUM']
    .fillna(df_rent['FLOOR_NUM'].median())
)

# BROKERAGE
df_rent['BROKERAGE'] = (
    df_rent['BROKERAGE']
    .fillna(0)
)

# Validation
print("Remaining Missing Values:")
print(df_rent.isnull().sum()[df_rent.isnull().sum() > 0])
```

```
Remaining Missing Values:
Series([], dtype: int64)
```

Missing Value Treatment (Rental Dataset)

Missing values in the rental dataset were handled using business-aware imputation techniques to maximize record retention while preserving the characteristics of the rental market.

- Price values were replaced using the median to retain rental listings while minimizing the influence of unusually high-end rental properties.
- Bedroom Count was imputed using the mode because bedroom configuration represents a discrete property characteristic and the most common value provides the most realistic replacement.
- Balcony Count was replaced with 0, assuming the information was unavailable or the property did not include a balcony.
- Floor Number was imputed using the median to preserve the distribution of apartment floor locations while reducing the impact of extreme values.
- Brokerage values were replaced with 0, treating missing brokerage information as unavailable or non-applicable.

Given the relatively small size of the rental dataset, the primary objective was to preserve as many valid listings as possible while maintaining analytical consistency.

Feature Engineering

```
for dataset in [df_sales, df_rent]:

    # Price Per Sq.Ft
    dataset['PRICE_SQFT'] = (
        dataset['PRICE'] / dataset['AREA']
    ).round(0)

    # BHK
    dataset.rename(
        columns={'BEDROOM_NUM': 'BHK'},
        inplace=True
    )


# Validation
print("Sales Dataset Shape :", df_sales.shape)
print("Rental Dataset Shape:", df_rent.shape)
```

```
Sales Dataset Shape : (10150, 21)
Rental Dataset Shape: (549, 21)
```

Feature Engineering

To improve analytical usability and create business-friendly metrics, two additional features were engineered for both the sales and rental datasets.

- Price per Sq.Ft (PRICE_SQFT) was calculated by dividing property price by area. This standardizes property valuation and enables fair comparison between properties of different sizes.
- BHK was created by renaming the bedroom count column to align with common real estate terminology used by buyers, renters, developers, and property platforms.

These engineered features serve as key inputs for pricing analysis, locality comparisons, value assessment, and dashboard development.

Data Quality Validation & Anomaly Treatment for Sales Dataset

```
print("Below ₹500 PPSF :", (df_sales['PRICE_SQFT'] < 500).sum())
print("Below ₹1000 PPSF:", (df_sales['PRICE_SQFT'] < 1000).sum())
print("Below ₹2000 PPSF:", (df_sales['PRICE_SQFT'] < 2000).sum())
```

```
Below ₹500 PPSF : 12
Below ₹1000 PPSF: 16
Below ₹2000 PPSF: 29
```

```
df_sales[df_sales['PRICE_SQFT'] < 1000][
    ['PROP_NAME','LOCALITY','PRICE','AREA','PRICE_SQFT','BHK']
].head(10)
```

```
                    PROP_NAME             LOCALITY         PRICE       AREA  \
129                 SOHNA ROAD                Sohna  3.000000e+08   784080.0
211                    Unknown             machroli  2.100000e+08   435600.0
244                    Unknown                Sohna  1.250000e+07    43560.0
245                    Unknown                Sohna  3.000000e+07    43560.0
805   Ansal Florence Residency  Sushant Lok Phase 3  2.100000e+07    24300.0
834                    Unknown    Sector 55 Gurgaon  2.000000e+07    22500.0
1320     Signature Global Park                Sohna  1.600000e+09  7840800.0
2358     Signature The Roselia   Sector 95A Gurgaon  2.500000e+07   514272.0
2918           Emaar Digihomes    Sector 62 Gurgaon  4.450000e+07   258879.0
3957     Signature The Roselia   Sector 95A Gurgaon  2.500000e+07   569243.0

      PRICE_SQFT  BHK
129        383.0  3.0
211        482.0  3.0
244        287.0  3.0
245        689.0  3.0
805        864.0  4.0
834        889.0  3.0
1320       204.0  3.0
2358        49.0  2.0
2918       172.0  3.0
3957        44.0  2.0
```

```
df_sales['AREA'].describe(percentiles=[0.01,0.05,0.10])
```

```
count    1.015000e+04
mean     3.624861e+03
std      7.948698e+04
min      3.000000e+01
1%       4.814900e+02
5%       9.000000e+02
10%      1.215000e+03
50%      2.081000e+03
max      7.840800e+06
Name: AREA, dtype: float64
```

```
print("Properties below 100 sq.ft :", (df['AREA'] < 100).sum())
print("Properties below 250 sq.ft :", (df['AREA'] < 250).sum())
print("Properties below 500 sq.ft :", (df['AREA'] < 500).sum())
print("Properties below 1000 sq.ft:", (df['AREA'] < 1000).sum())
```

```
Properties below 100 sq.ft : 1
Properties below 250 sq.ft : 28
Properties below 500 sq.ft : 123
Properties below 1000 sq.ft: 696
```

```
df_sales[df_sales['AREA'] > 50000][
    ['PROP_NAME','LOCALITY','PRICE','AREA','PRICE_SQFT']
].sort_values('AREA', ascending=False)
```

```
                     PROP_NAME             LOCALITY         PRICE       AREA  \
1320      Signature Global Park                Sohna  1.600000e+09  7840800.0
129                  SOHNA ROAD                Sohna  3.000000e+08   784080.0
9832  Signature Global Solera 2  Sector 107 Gurgaon  2.500000e+07   577946.0
3958      Signature The Roselia  Sector 95A Gurgaon  2.500000e+07   569243.0
3957      Signature The Roselia  Sector 95A Gurgaon  2.500000e+07   569243.0
9827  Signature Global Solera 2  Sector 107 Gurgaon  2.500000e+07   553856.0
3959      Signature The Roselia  Sector 95A Gurgaon  2.500000e+07   514272.0
2358      Signature The Roselia  Sector 95A Gurgaon  2.500000e+07   514272.0
211                      Unknown             machroli  2.100000e+08   435600.0
2918            Emaar Digihomes   Sector 62 Gurgaon  4.450000e+07   258879.0
727                      Unknown  Sector 114 Gurgaon  3.000000e+08    58500.0
4023      Pyramid Urban Homes 2   Sector 86 Gurgaon  2.500000e+07    51557.0

      PRICE_SQFT
1320       204.0
```

```
129        383.0
9832        43.0
3958        44.0
3957        44.0
9827        45.0
3959        49.0
2358        49.0
211        482.0
2918       172.0
727       5128.0
4023       485.0
```

```
original_records = len(df_sales)

df_sales = df_sales[
    (df_sales['AREA'] >= 250) &
    (df_sales['AREA'] <= 50000)
].copy()

print("Original Records :", original_records)
print("Cleaned Records  :", len(df_sales))
print("Records Removed  :", original_records - len(df_sales))
print(
    "Percentage Removed:",
    round(
        ((original_records - len(df_sales)) / original_records) * 100,
        2
    ),
    "%"
)
```

```
Original Records : 10150
Cleaned Records  : 10112
Records Removed  : 38
Percentage Removed: 0.37 %
```

```
df_sales['PRICE_SQFT'] = (
    df_sales['PRICE'] / df_sales['AREA']
).round(0)

df_sales['PRICE_SQFT'].describe(
    percentiles=[0.90,0.95,0.99,0.995,0.999]
)
```

```
count    1.011200e+04
mean     1.574043e+04
std      2.205432e+04
min      2.870000e+02
50%      1.229500e+04
90%      2.555600e+04
95%      3.468735e+04
99%      5.555600e+04
99.5%    7.142900e+04
99.9%    2.737411e+05
max      1.193416e+06
Name: PRICE_SQFT, dtype: float64
```

```
df_sales.nsmallest(
    20,
    'PRICE_SQFT'
)[
    ['PROP_NAME','LOCALITY','PRICE','AREA','PRICE_SQFT','BHK']
]
```

```
                                PROP_NAME             LOCALITY       PRICE  \
244                               Unknown                Sohna  12500000.0
245                               Unknown                Sohna  30000000.0
9794           Godrej Nature Plus Serenity      Sector-33 Sohna  11700000.0
805               Ansal Florence Residency  Sushant Lok Phase 3  21000000.0
834                               Unknown    Sector 55 Gurgaon  20000000.0
1694                          Vatika City    Sector 49 Gurgaon  30000000.0
8517   Shapoorji Pallonji Joyville Gurugram   Sector 102 Gurgaon  20200000.0
10002                    Puri Emerald Bay   Sector 104 Gurgaon  30600000.0
3541                              Unknown          Palam Vihar  30000000.0
5487          Sushant Lok 1 Builder Floors  Sushant Lok Phase 1  35000000.0
1802                DLF City Plots Phase 4          DLF Phase 4  32500000.0
1801                DLF City Plots Phase 2          DLF Phase 2  24800000.0
3518                          Vatika City    Sector 49 Gurgaon  20000000.0
2721                              Unknown          Palam Vihar  23000000.0
5939                   Ambience Caitriona          DLF Phase 3  25000000.0
6244           Adani Brahma Samsara Vilasa    Sector 63 Gurgaon  42000000.0
4863                     DLF The Camellias    Sector 42 Gurgaon  25000000.0
1174                              Unknown          DLF Phase 4  85000000.0
5933                     DLF The Camellias    Sector 42 Gurgaon  25000000.0
4864                       DLF The Aralias    Sector 42 Gurgaon  25000000.0

          AREA  PRICE_SQFT  BHK
```

```
244    43560.0       287.0  3.0
245    43560.0       689.0  3.0
9794   15123.0       774.0  2.0
805    24300.0       864.0  4.0
834    22500.0       889.0  3.0
1694   26602.0      1128.0  4.0
8517   16668.0      1212.0  3.0
10002  22450.0      1363.0  3.0
3541   21870.0      1372.0  5.0
5487   24300.0      1440.0  4.0
1802   21870.0      1486.0  3.0
1801   16200.0      1531.0  3.0
3518   12809.0      1561.0  3.0
2721   14013.0      1641.0  3.0
5939   15000.0      1667.0  7.0
6244   22500.0      1867.0  4.0
4863   13000.0      1923.0  5.0
1174   43560.0      1951.0  3.0
5933   11000.0      2273.0  6.0
4864   10000.0      2500.0  5.0
```

```
df_sales['PRICE_SQFT'].describe(
    percentiles=[0.01,0.02,0.05,0.10]
)
```

```
count    1.011200e+04
mean     1.574043e+04
std      2.205432e+04
min      2.870000e+02
1%       5.221990e+03
2%       5.929100e+03
5%       6.803200e+03
10%      7.576000e+03
50%      1.229500e+04
max      1.193416e+06
Name: PRICE_SQFT, dtype: float64
```

```
print((df_sales['PRICE_SQFT'] < 5000).sum())
```

```
78
```

```
ppsf_cutoff = df_sales['PRICE_SQFT'].quantile(0.01)

df_sales = df_sales[
    df_sales['PRICE_SQFT'] >= ppsf_cutoff
].copy()

print("PPSF Threshold:", round(ppsf_cutoff, 0))
print("Records Remaining:", len(df_sales))
```

```
PPSF Threshold: 5222.0
Records Remaining: 10010
```

Data Quality Validation & Anomaly Treatment

Before commencing exploratory analysis, the sales dataset was subjected to a final data-quality validation process to identify and remove records that could distort pricing and valuation insights.

Area-Based Validation

Property area is a fundamental driver of valuation and directly influences the calculation of price per square foot. During validation, a small number of records were identified with unusually low or exceptionally large area values that were inconsistent with typical residential properties.

To improve analytical reliability, properties were retained only if:

- Area ≥ 250 sq.ft
- Area ≤ 50,000 sq.ft

The lower threshold removes extremely small records that are unlikely to represent standard residential properties, while the upper threshold eliminates unusually large listings that were found to generate unrealistic valuation metrics and likely reflect data-entry, unit-conversion, or scraping anomalies.

Price-per-Square-Foot Validation

After correcting area-related anomalies, the distribution of Price per Sq.ft (PPSF) was analyzed to identify properties with unusually low valuations relative to the broader Gurgaon residential market.

Rather than applying an arbitrary cutoff, a percentile-based approach was adopted. Properties below the 1st percentile of the PPSF distribution (₹5,222 per sq.ft) were treated as anomalous observations and removed from the analytical dataset.

This approach ensures that only the most extreme lower-tail observations are excluded while preserving the genuine diversity of Gurgaon's housing market.

Impact of Cleaning

- Original Sales Records: 10,150
- Final Analytical Records: 10,010
- Records Removed: 140
- Percentage Removed: ~1.38%

The cleaning process affected only a small fraction of the dataset while significantly improving the reliability of pricing, locality, and valuation analyses. The resulting dataset provides a robust foundation for exploratory analysis, investment assessment, and business recommendations.

Exploratory Data Analysis (EDA)

Sales Market

Following data cleaning, feature engineering, and market segmentation, the analysis now focuses exclusively on the residential sales market. Rental listings were separated to ensure that pricing patterns, valuation metrics, and market dynamics are not distorted by fundamentally different rental pricing behavior.

The objective of this exploratory analysis is to uncover key pricing trends, understand the factors influencing property valuations, identify locality-level opportunities, and generate actionable insights for homebuyers and real estate investors in Gurgaon.

Objective

To uncover:

- Pricing patterns
- Locality trends
- Key drivers of property value
- Investment opportunities

```
print(f"Total Properties: {df_sales.shape[0]}")
print(f"Total Localities: {df_sales['LOCALITY'].nunique()}")
print(f"Average Price: ₹{round(df_sales['PRICE'].mean(), 2)}")
print(f"Average Area: {round(df_sales['AREA'].mean(), 0)} sq.ft")
print(f"Average Price per Sq.ft: ₹{round(df_sales['PRICE_SQFT'].mean(), 0)}")
print(f"Median Price: ₹{round(df_sales['PRICE'].median(),2)}")
print(f"Median Area: {round(df_sales['AREA'].median(),0)} sq.ft")
print(f"Median Price per Sq.ft: ₹{round(df_sales['PRICE_SQFT'].median(),0)}")
```

```
Total Properties: 10010
Total Localities: 222
Average Price: ₹35487562.44
Average Area: 2325.0 sq.ft
Average Price per Sq.ft: ₹15861.0
Median Price: ₹25000000.0
Median Area: 2070.0 sq.ft
Median Price per Sq.ft: ₹12382.0
```

Key Insights

- The Gurgaon housing market consists of 10,010 residential properties for sale across 222 localities, providing broad coverage of the city's residential landscape and enabling meaningful locality-level analysis.
- The market is predominantly centered around the mid-segment, with a median property price of ₹2.5 Cr and a median property size of 2,070 sq.ft, indicating that most listings target upper-middle-income homebuyers rather than luxury buyers.

- While the median price-per-square-foot stands at ₹12,382, the significantly higher average of ₹15,861 suggests the presence of premium micro-markets that command substantial pricing premiums and pull the overall market average upward.

Pricing Landscape

Understanding Property Valuation Across Gurgaon

Price per square foot (PPSF) is one of the most important metrics in real estate analysis as it standardizes property valuation across homes of different sizes. While total property price is influenced by factors such as area, configuration, and amenities, PPSF provides a more consistent measure for comparing the relative value of properties across the market.

In this section, we examine the distribution of price per square foot across Gurgaon to understand overall market valuation patterns. The analysis also helps identify the presence of unusually expensive properties or localities that may influence market-level statistics and provides an initial indication of potential outliers that require further investigation.

```
import matplotlib.pyplot as plt
import seaborn as sns

# Statistics
mean_ppsf = df_sales['PRICE_SQFT'].mean()
median_ppsf = df_sales['PRICE_SQFT'].median()

fig, ax = plt.subplots(1, 2, figsize=(14,5))

# Histogram
sns.histplot(
    df_sales['PRICE_SQFT'],
    bins=50,
    color='#C8D5B9',
    kde=True,
    ax=ax[0]
)

ax[0].axvline(
    mean_ppsf,
    linestyle='--',
    linewidth=2,
    color='#D98C7A',
    label=f'Mean: ₹{mean_ppsf:,.0f}'
)

ax[0].axvline(
    median_ppsf,
    linestyle=':',
    linewidth=2,
    label=f'Median: ₹{median_ppsf:,.0f}'
)

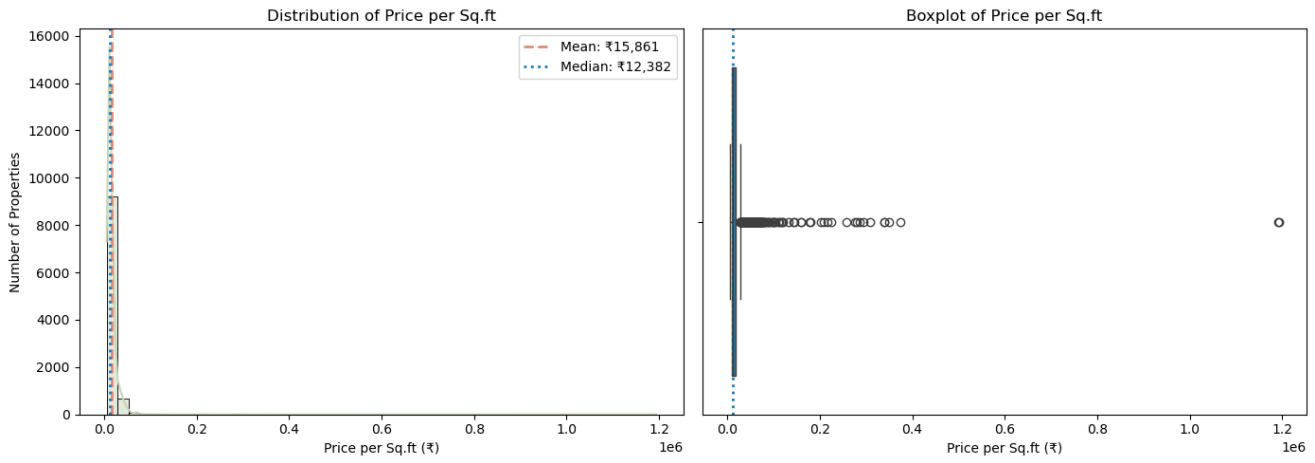
ax[0].set_title('Distribution of Price per Sq.ft')
ax[0].set_xlabel('Price per Sq.ft (₹)')
ax[0].set_ylabel('Number of Properties')
ax[0].legend()

# Boxplot
sns.boxplot(
    x=df_sales['PRICE_SQFT'],
    ax=ax[1]
)

ax[1].axvline(
    median_ppsf,
    linestyle=':',
    linewidth=2
)

ax[1].set_title('Boxplot of Price per Sq.ft')
ax[1].set_xlabel('Price per Sq.ft (₹)')

plt.tight_layout()
plt.show()
```



Key Insights

- The distribution of price per square foot remains positively skewed, with most properties concentrated within a relatively narrow valuation range while a smaller number of premium properties command substantially higher PPSF values.
- The median PPSF of ₹12,382 compared with the mean of ₹15,861 indicates the presence of premium micro-markets, though the gap is moderate and reflects a more realistic market structure than previously observed.
- The boxplot highlights a small number of extreme high-value properties, suggesting that Gurgaon contains distinct luxury segments that operate at pricing levels significantly above the broader residential market.

```
df_sales[['PROP_ID','LOCALITY','PRICE','AREA','PRICE_SQFT']]\
.sort_values('PRICE_SQFT', ascending=False)\
.head()
```

	PROP_ID	LOCALITY	PRICE	AREA	PRICE_SQFT
9209	M71183824	DLF Phase 3	2.900000e+09	2430.0	1193416.0
4952	T71573082	Sector 92 Gurgaon	2.250000e+09	1890.0	1190476.0
7407	N70971812	Sector 66 Gurgaon	1.875000e+08	500.0	375000.0
553	P71453590	Sector 26 Gurgaon	1.750000e+08	500.0	350000.0
5327	V67302720	Sector 66 Gurgaon	1.700000e+08	500.0	340000.0

```
df_sales['PRICE_SQFT'].describe(
    percentiles=[0.90,0.95,0.99,0.995,0.999]
)
```

```
count    1.001000e+04
mean     1.586134e+04
std      2.213325e+04
min      5.230000e+03
50%      1.238250e+04
90%      2.571450e+04
95%      3.493850e+04
99%      5.555600e+04
99.5%    7.142900e+04
99.9%    2.755090e+05
max      1.193416e+06
Name: PRICE_SQFT, dtype: float64
```

Key Insights

- The typical residential property in Gurgaon is valued at approximately ₹12,300 per sq.ft, indicating that most of the market remains concentrated within a relatively narrow valuation range despite significant locality-level differences.
- The gap between the median (₹12,382) and mean (₹15,861) price per sq.ft reveals a right-skewed market, where a relatively small number of premium properties exert a disproportionate influence on overall market averages.
- Property valuations increase sharply in the upper tail of the market, with the top 5% of properties exceeding ₹35,000 per sq.ft and the top 1% crossing ₹55,000 per sq.ft, highlighting the presence of a distinct luxury segment within Gurgaon’s residential market.

Outlier Investigation & Treatment

Understanding Extreme Property Valuations

The previous analysis revealed a highly skewed distribution of price per square foot, with a small number of observations extending far beyond the typical market range. Such observations can significantly distort summary statistics, visualizations, and downstream analyses.

However, not all outliers should be removed. In real estate datasets, extreme values may represent genuine luxury properties, unique market segments, data-entry errors, or inconsistencies introduced during data collection.

The objective of this section is to identify the extent of these extreme observations, investigate their characteristics, and determine an appropriate treatment strategy before proceeding with further market analysis.

```
p99 = df_sales['PRICE_SQFT'].quantile(0.99)
p995 = df_sales['PRICE_SQFT'].quantile(0.995)
p999 = df_sales['PRICE_SQFT'].quantile(0.999)

print(f"Properties above 99th percentile: {(df_sales['PRICE_SQFT'] > p99).sum()}")
print(f"Properties above 99.5th percentile: {(df_sales['PRICE_SQFT'] > p995).sum()}")
print(f"Properties above 99.9th percentile: {(df_sales['PRICE_SQFT'] > p999).sum()}")
```

```
Properties above 99th percentile: 99
Properties above 99.5th percentile: 48
Properties above 99.9th percentile: 11
```

```
percentiles = {
    '50th': df_sales['PRICE_SQFT'].quantile(0.50),
    '75th': df_sales['PRICE_SQFT'].quantile(0.75),
    '90th': df_sales['PRICE_SQFT'].quantile(0.90),
    '95th': df_sales['PRICE_SQFT'].quantile(0.95),
    '99th': df_sales['PRICE_SQFT'].quantile(0.99),
    '99.5th': df_sales['PRICE_SQFT'].quantile(0.995),
    '99.9th': df_sales['PRICE_SQFT'].quantile(0.999)
}


plt.figure(figsize=(10, 5))

bars = plt.bar(
    percentiles.keys(),
    percentiles.values(),
    color='#C8D5B9',      # Sophisticated pastel mint
    edgecolor='black',
    linewidth=0.8
)

# Title & Labels
plt.title(
    'Price per Sq.ft Across Key Percentiles',
    fontsize=14,
    pad=15
)

plt.ylabel('Price per Sq.ft (₹)', fontsize=11)

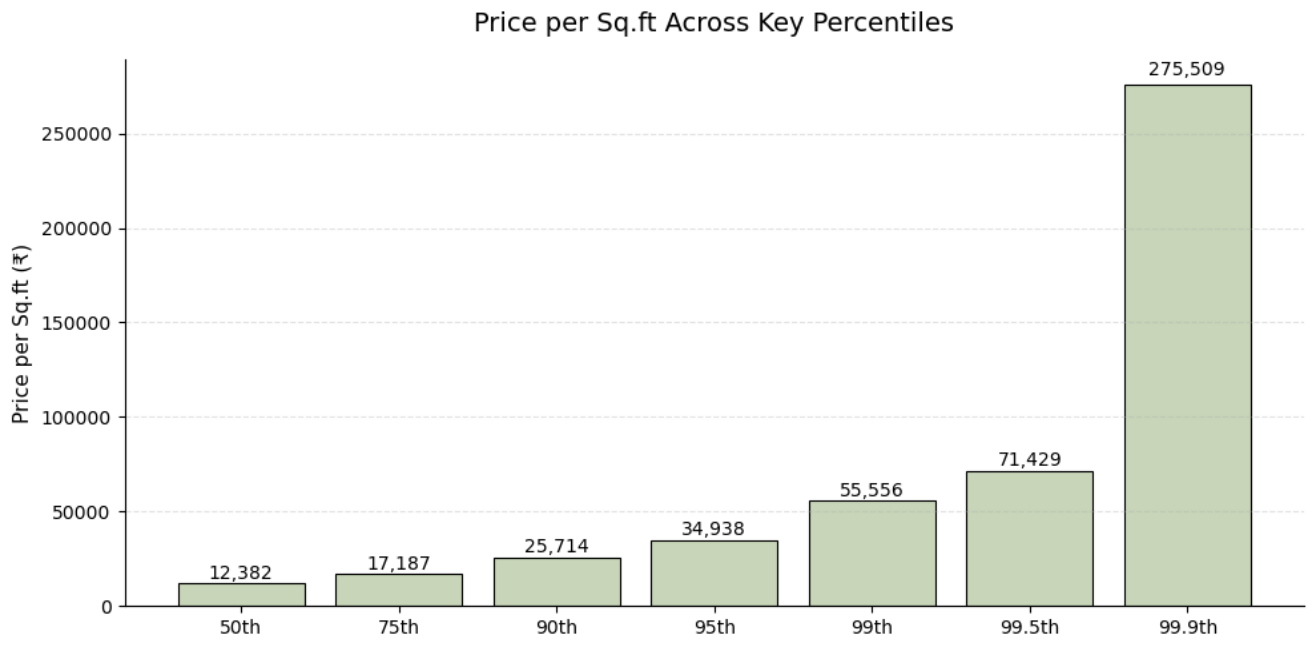
# Vertical Gridlines
plt.grid(
    axis='y',
    linestyle='--',
    alpha=0.35
)

# Remove top and right spines
ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

# Value Labels
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + (height * 0.01),
        f'{height:,.0f}',
        ha='center',
        va='bottom',
```

```
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



Key Insights

- Gurgaon’s residential market remains relatively stable through most of the distribution, with the median property priced at ₹12,381 per sq.ft and even the 95th percentile remaining below ₹35,000 per sq.ft, indicating that extreme valuations are concentrated in a small segment of the market.
- Property valuations accelerate sharply beyond the 99th percentile, where price per sq.ft rises from ₹55,556 at the 99th percentile to over ₹71,400 at the 99.5th percentile, highlighting the presence of a distinct ultra-premium housing segment.
- Only 99 properties (1% of listings) exceed the 99th percentile threshold, while just 11 properties exceed the 99.9th percentile threshold, confirming that the most extreme price-per-square-foot values are highly concentrated and should be investigated separately as either genuine luxury assets or potential listing anomalies.

Market Structure Analysis

Understanding Housing Supply Composition

Before analyzing pricing dynamics and locality-level performance, it is important to understand the composition of the housing inventory itself. The distribution of BHK configurations provides insight into the type of residential demand being served by the market and helps identify whether Gurgaon is primarily a mass-market, family-oriented, or luxury housing destination.

The following analysis examines the distribution of available property listings across different BHK categories to better understand the structure of the Gurgaon residential market.

```
import matplotlib.pyplot as plt

bhk_counts = (
    df_sales['BHK']
    .value_counts()
    .sort_index()
)

plt.figure(figsize=(10,5))

bars = plt.bar(
    bhk_counts.index.astype(str),
    bhk_counts.values,
    color='#C8D5B9',
    edgecolor='black',
```

```
    linewidth=0.8
)

plt.title(
    'Distribution of Properties by BHK Configuration',
    fontsize=14,
    pad=15
)

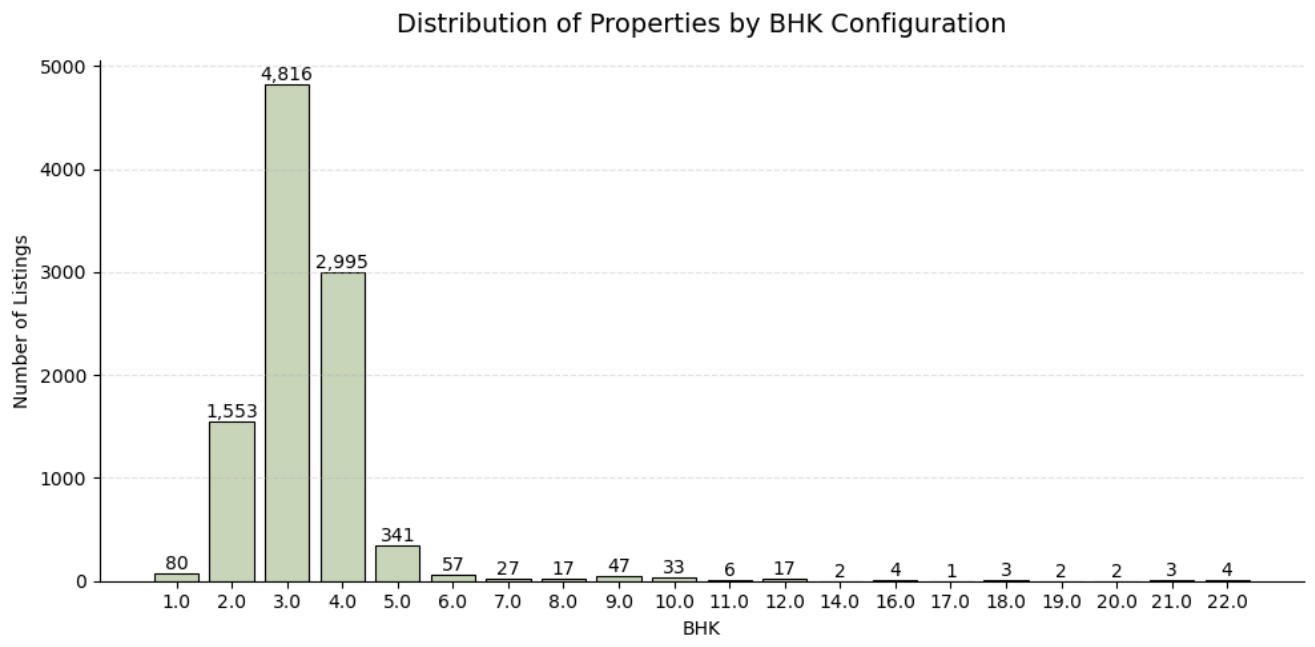
plt.xlabel('BHK')
plt.ylabel('Number of Listings')

plt.grid(
    axis='y',
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

# Count Labels
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height,
        f'{height:,}',
        ha='center',
        va='bottom',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```




```
(
    df_sales['BHK']
    .value_counts(normalize=True)
    .mul(100)
    .round(1)
    .sort_index()
)
```

```
BHK
1.0      0.8
2.0     15.5
3.0     48.1
4.0     29.9
5.0      3.4
6.0      0.6
7.0      0.3
8.0      0.2
9.0      0.5
10.0     0.3
```

```
11.0     0.1
12.0     0.2
14.0     0.0
16.0     0.0
17.0     0.0
18.0     0.0
19.0     0.0
20.0     0.0
21.0     0.0
22.0     0.0
Name: proportion, dtype: float64
```

```
df_sales[df_sales['BHK'] > 10][
    ['LOCALITY','BHK','AREA','PRICE']
].sort_values('BHK', ascending=False).head(10)
```

	LOCALITY	BHK	AREA	PRICE
9393	C Block Sushant Lok Phase 1	22.0	1935.0	100000000.0
9432	C Block Sushant Lok Phase 1	22.0	1080.0	65000000.0
9545	Sushant Lok Phase 1	22.0	5400.0	54500000.0
6774	South City 2	22.0	540.0	42500000.0
9376	C Block Sushant Lok Phase 1	21.0	1080.0	62500000.0
9546	Sushant Lok Phase 1	21.0	5400.0	55000000.0
9372	Sushant Lok Phase 1	21.0	1080.0	65000000.0
4925	Saraswati Vihar	20.0	1800.0	75000000.0
9415	C Block Sushant Lok Phase 1	20.0	1080.0	65000000.0
101	Sector 42 Gurgaon	19.0	512.0	103500000.0

```
df_sales['BHK_GROUP'] = df_sales['BHK'].apply(
    lambda x: '6+ BHK' if x >= 6 else f'{int(x)} BHK'
)

import matplotlib.pyplot as plt

# Desired order
order = ['1 BHK', '2 BHK', '3 BHK', '4 BHK', '5 BHK', '6+ BHK']

# Percentage distribution
bhk_pct = (
    df_sales['BHK_GROUP']
    .value_counts(normalize=True)
    .mul(100)
    .reindex(order)
)

plt.figure(figsize=(10,5))

bars = plt.bar(
    bhk_pct.index,
    bhk_pct.values,
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

# Title
plt.title(
    'Housing Inventory Distribution by BHK Category',
    fontsize=14,
    pad=15
)

plt.ylabel('Share of Listings (%)', fontsize=11)

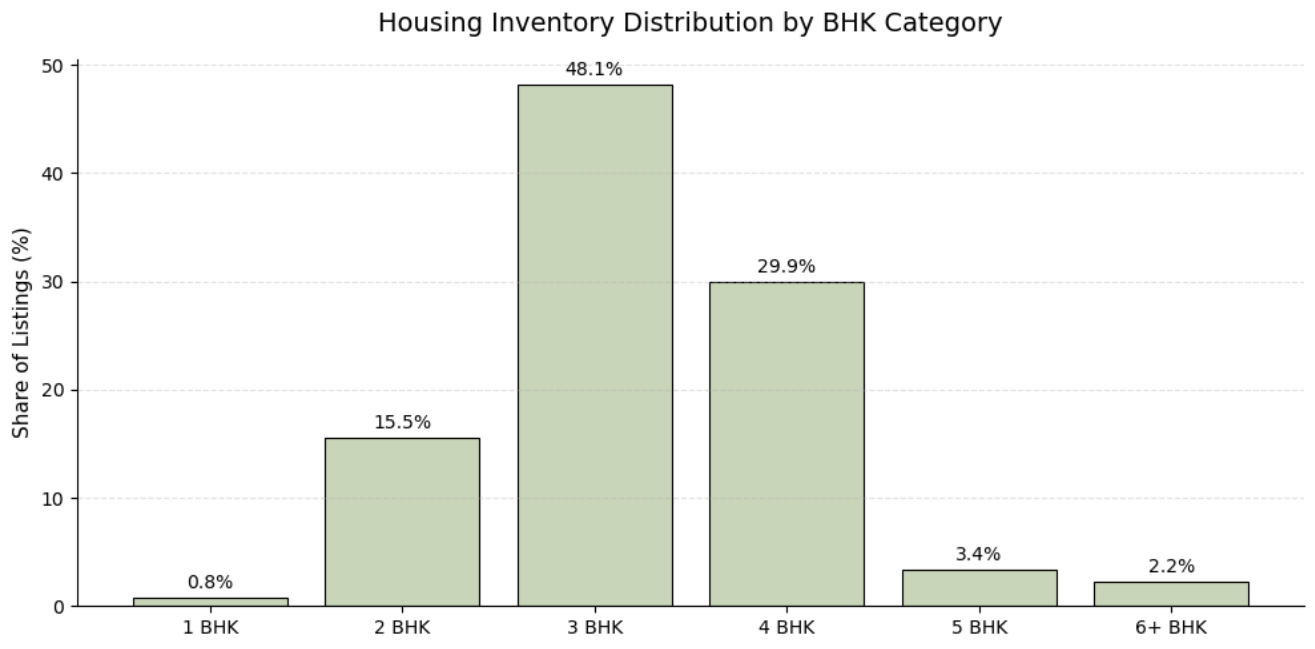
# Grid
plt.grid(
    axis='y',
    linestyle='--',
    alpha=0.35
)

# Clean spines
ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

# Percentage labels
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + 0.5,
        f'{height:.1f}%',
        ha='center',
        va='bottom',
```

```
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



Key Insights

- Gurgaon is predominantly a family-oriented housing market, with 3 BHK and 4 BHK properties accounting for over 78% of total listings, making them the dominant housing configurations across the city.
- The 3 BHK segment alone represents nearly half of all available inventory (48.1%), indicating strong alignment between market supply and demand from mid-to-upper-income households.
- Entry-level housing remains limited, with 1 BHK properties contributing less than 1% of listings, suggesting that Gurgaon is primarily positioned as a premium residential market rather than a starter-home destination.

Understanding Property Price Distribution

While the BHK analysis reveals the composition of housing inventory, it does not explain how property values are distributed across the market. Understanding the distribution of property prices helps identify whether Gurgaon is characterized by a concentrated pricing range or a wide spectrum of housing options.

The following analysis examines the distribution of residential property prices to better understand the overall pricing structure of the market.

```
mean_price = df_sales['PRICE'].mean()
median_price = df_sales['PRICE'].median()

plt.figure(figsize=(10,5))

sns.histplot(
    df_sales['PRICE'],
    color='#C8D5B9',
    bins=40,
    kde=True
)

plt.axvline(
    mean_price,
    linestyle='--',
    color='#D98C7A',
    linewidth=2,
    label=f'Mean: ₹{mean_price:,.0f}'
)

plt.axvline(
```

```
    median_price,
    color='#7FA8A1',
    linestyle='--',
    linewidth=2,
    label=f'Median: ₹{median_price:,.0f}'
)

plt.title(
    'Distribution of Property Prices',
    fontsize=14,
    pad=15
)

plt.xlabel('Property Price')
plt.ylabel('Number of Listings')

plt.legend()

plt.grid(
    axis='y',
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.tight_layout()
plt.show()
```




```
df_sales['PRICE'].describe(
    percentiles=[0.25,0.50,0.75,0.90,0.95,0.99]
)
```

```
count    1.001000e+04
mean     3.548756e+07
std      5.355517e+07
min      1.000000e+07
25%      1.800000e+07
50%      2.500000e+07
75%      3.800000e+07
90%      6.250000e+07
95%      8.720000e+07
99%      1.800000e+08
max      2.900000e+09
Name: PRICE, dtype: float64
```

Key Insights

- Property prices in Gurgaon exhibit a strongly right-skewed distribution, with a median price of ₹2.5 Cr and a substantially higher mean price of ₹3.5 Cr, indicating the presence of premium and luxury properties that elevate the market average.

- The majority of residential listings remain concentrated within the mid-market segment, with 75% of properties priced below ₹3.8 Cr and 90% below ₹6.2 Cr, highlighting the dominance of upper-middle-income housing inventory.
- While premium properties exist throughout the market, the luxury segment remains relatively small, as only 1% of listings exceed ₹18 Cr, demonstrating that extreme high-value properties represent a niche portion of Gurgaon's housing landscape.

Locality Intelligence

Understanding Inventory Concentration Across Localities

Location is one of the most influential factors in residential real estate. Before comparing pricing, property sizes, or investment potential across localities, it is important to understand where housing supply is concentrated.

Analyzing listing volumes by locality helps identify the major residential hubs of Gurgaon and highlights the areas contributing the largest share of available inventory. These localities are likely to play a significant role in shaping overall market trends and warrant deeper investigation in subsequent analyses.

```
import matplotlib.pyplot as plt

top_localities = (
    df_sales['LOCALITY']
    .value_counts()
    .head(10)
    .sort_values()
)

plt.figure(figsize=(10,6))

bars = plt.barh(
    top_localities.index,
    top_localities.values,
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

plt.title(
    'Top 10 Localities by Listing Volume',
    fontsize=14,
    pad=15
)

plt.xlabel('Number of Listings')

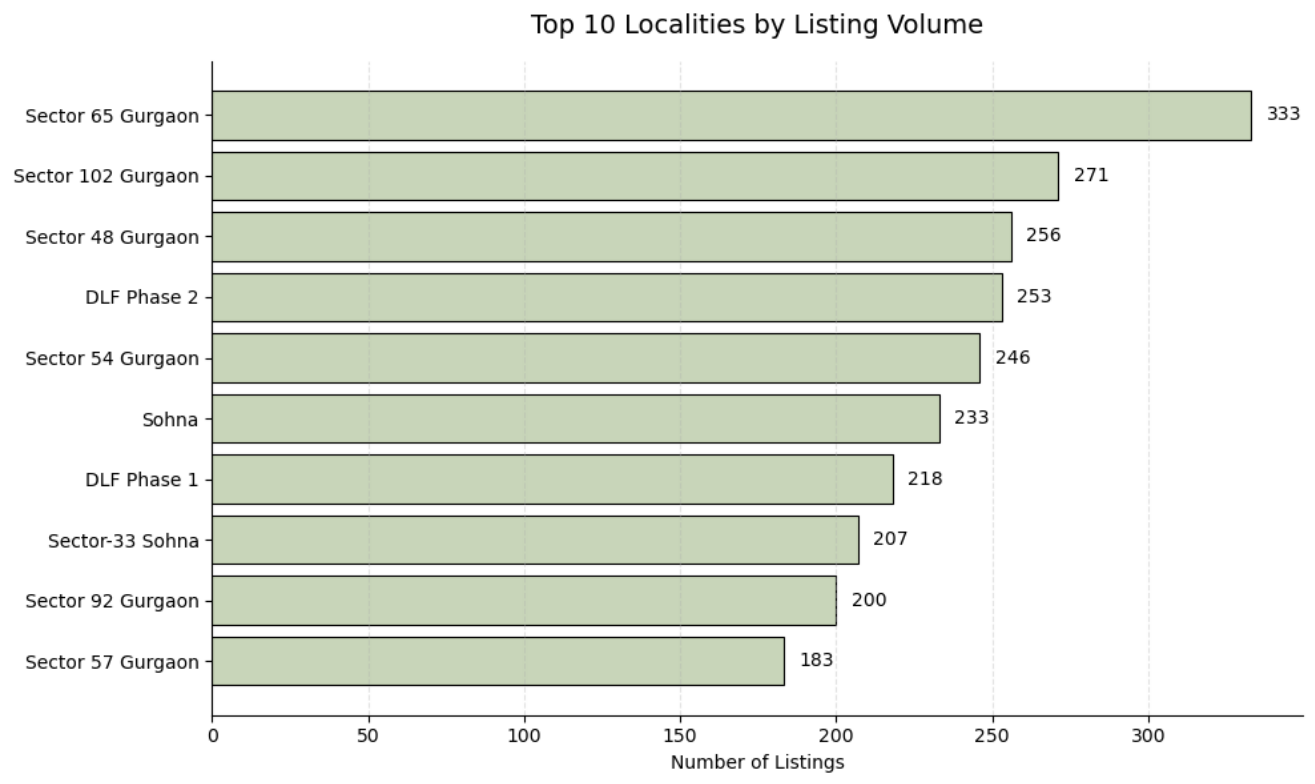
plt.grid(
    axis='x',
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

for bar in bars:
    width = bar.get_width()

    plt.text(
        width + 5,
        bar.get_y() + bar.get_height()/2,
        f'{int(width):,}',
        va='center',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



Key Insights

- Housing inventory in Gurgaon is concentrated within a relatively small number of micro-markets, with Sector 65 Gurgaon leading the market at 333 active listings, followed by Sector 102 Gurgaon, Sector 48 Gurgaon, and DLF Phase 2.
- The presence of multiple established sectors and DLF localities among the top-ranked markets suggests that buyer activity and housing supply remain concentrated in well-developed residential corridors rather than being evenly distributed across the city.
- The gap between the highest-ranked locality (333 listings) and the tenth-ranked locality (183 listings) is relatively modest, indicating that Gurgaon's residential inventory is spread across several active markets rather than being dominated by a single locality.

Identifying High-Value Residential Markets

Listing volume indicates where housing supply is concentrated, but it does not necessarily reflect market value. To understand which areas command the strongest valuations, it is useful to compare localities using average price per square foot.

Price per square foot provides a standardized measure of property value and enables meaningful comparison across localities with different property sizes. The following analysis highlights the highest-valued residential markets within Gurgaon.

```
locality_stats = (
    df_sales.groupby('LOCALITY')
    .agg(
        listings=('LOCALITY', 'count'),
        median_ppsf=('PRICE_SQFT', 'median')
    )
    .reset_index()
)

# Minimum listing threshold
locality_stats = locality_stats[
    locality_stats['listings'] >= 20
]

top_localities = (
    locality_stats
    .sort_values('median_ppsf', ascending=False)
    .head(10)
    .sort_values('median_ppsf')
)
```

```
plt.figure(figsize=(10,6))

bars = plt.barh(
    top_localities['LOCALITY'],
    top_localities['median_ppsf'],
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

plt.title(
    'Top 10 Localities by Median Price per Sq.ft',
    fontsize=14,
    pad=15
)

plt.xlabel('Median Price per Sq.ft (₹)')

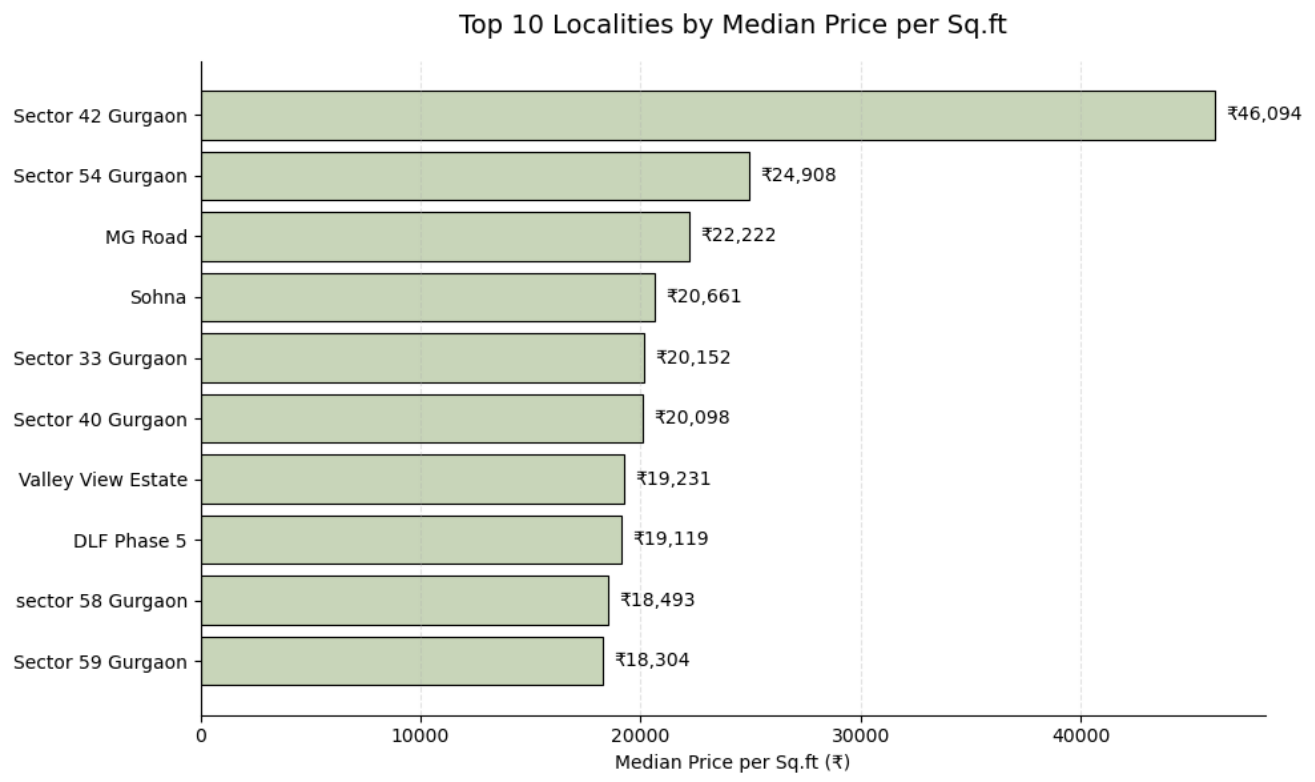
plt.grid(
    axis='x',
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

for bar in bars:
    width = bar.get_width()

    plt.text(
        width + 500,
        bar.get_y() + bar.get_height()/2,
        f'₹{width:,.0f}',
        va='center',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```




```
df_sales[df_sales['LOCALITY'] == 'Sector 42 Gurgaon'].shape[0]
```

```
95
```

```
df_sales[df_sales['LOCALITY'] == 'Sector 42 Gurgaon'][
    ['PRICE','AREA','PRICE_SQFT']
].describe()
```

```
              PRICE          AREA      PRICE_SQFT
```

```
count  8.300000e+01     83.000000      83.000000
mean   2.916024e+08   6012.927711   45565.072289
std    1.966382e+08   2399.335496   23945.540136
min    1.200000e+07    512.000000    9302.000000
25%    2.400000e+08   5800.000000   41028.500000
50%    2.925000e+08   6400.000000   46094.000000
75%    3.450000e+08   6401.000000   51516.500000
max    1.500000e+09  17000.000000  202148.000000
```

Key Insights

- Sector 42 Gurgaon clearly dominates Gurgaon's premium housing market, with a median price of approximately ₹46,000 per sq.ft—nearly 85% higher than the second-ranked locality. This indicates that buyers are paying a substantial location premium driven by exclusivity, connectivity, and luxury positioning rather than property size alone.
- Premium pricing is highly concentrated in a small number of micro-markets. While most top localities fall within the ₹18,000–₹25,000 per sq.ft range, Sector 42 Gurgaon stands out as a distinct outlier, highlighting the presence of a separate ultra-luxury segment within Gurgaon's residential market.
- Despite its premium valuation, Sector 42 Gurgaon also contains relatively large properties, with a median property size of approximately 6,400 sq.ft and a median property price of ₹29.25 Cr. This suggests that its exceptional price-per-square-foot levels are not driven by smaller luxury units but by strong location-driven demand and high-end property offerings.

Understanding the Relationship Between Area and Property Price

Property size is one of the most important factors influencing residential real estate valuations. Larger properties typically command higher prices; however, the strength of this relationship varies across markets depending on location, property quality, and buyer preferences.

The following analysis examines how property prices change with area and helps determine whether size is a primary driver of value within the Gurgaon housing market.

```
plot_df = df_sales[
    df_sales['AREA'] <= df_sales['AREA'].quantile(0.99)
]

import seaborn as sns
import matplotlib.pyplot as plt

# Sample for visualization
sample_df = plot_df.sample(
    n=min(3000, len(df_sales)),
    random_state=42
)

plt.figure(figsize=(10,6))

sns.regplot(
    data=sample_df,
    x='AREA',
    y='PRICE',
    color='#6F9690',
    scatter_kws={
        'alpha': 0.35,
        's': 25
    },
    line_kws={
        'linewidth': 2
    }
)

plt.title(
    'Relationship Between Property Area and Price',
    fontsize=14,
    pad=15
)

plt.xlabel('Area (Sq.ft)')
plt.ylabel('Property Price (₹)')

plt.grid(
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
```

```
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.tight_layout()
plt.show()
```




```
corr = df_sales['AREA'].corr(df_sales['PRICE'])

print(f"Correlation between Area and Price: {corr:.2f}")
```

```
Correlation between Area and Price: 0.48
```

Key Insights

- Property size remains one of the strongest drivers of residential property prices in Gurgaon, with a moderate positive correlation of 0.48 between area and price. The increase in correlation after anomaly treatment suggests that data-quality issues were previously masking part of the true relationship.
- Larger properties generally command higher prices, but the relationship is far from perfect. The wide spread of prices among similarly sized homes indicates that factors such as locality, project quality, amenities, and market positioning also play a significant role in determining property values.
- The scatterplot reveals the presence of premium micro-markets where properties achieve substantially higher valuations despite having comparable sizes. This highlights that Gurgaon’s residential market is influenced not only by property size but also by strong location-driven pricing premiums.

Locality Positioning Analysis

Residential markets are not defined solely by pricing or inventory levels. Some localities attract significant housing supply due to broad market demand, while others command premium valuations despite having relatively limited inventory.

By comparing listing volume and median price per square foot simultaneously, we can identify different types of residential markets, including established premium hubs, high-volume mass-market locations, and niche luxury areas. This provides a more holistic view of Gurgaon's residential landscape than ranking localities on a single metric alone.

```
locality_map = (
    df_sales.groupby('LOCALITY')
    .agg(
        listings=('LOCALITY', 'count'),
        median_ppsf=('PRICE_SQFT', 'median')
    )
    .reset_index()
```

```
)

# Keep meaningful markets
locality_map = locality_map[
    locality_map['listings'] >= 15
]

import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,6))

sns.scatterplot(
    data=locality_map,
    x='listings',
    y='median_ppsf',
    s=80,
    alpha=0.7,
    color='#9FB29A'
)

# Median reference lines
plt.axvline(
    locality_map['listings'].median(),
    linestyle='--',
    alpha=0.6
)

plt.axhline(
    locality_map['median_ppsf'].median(),
    linestyle='--',
    color="#5F8781",
    alpha=0.6
)

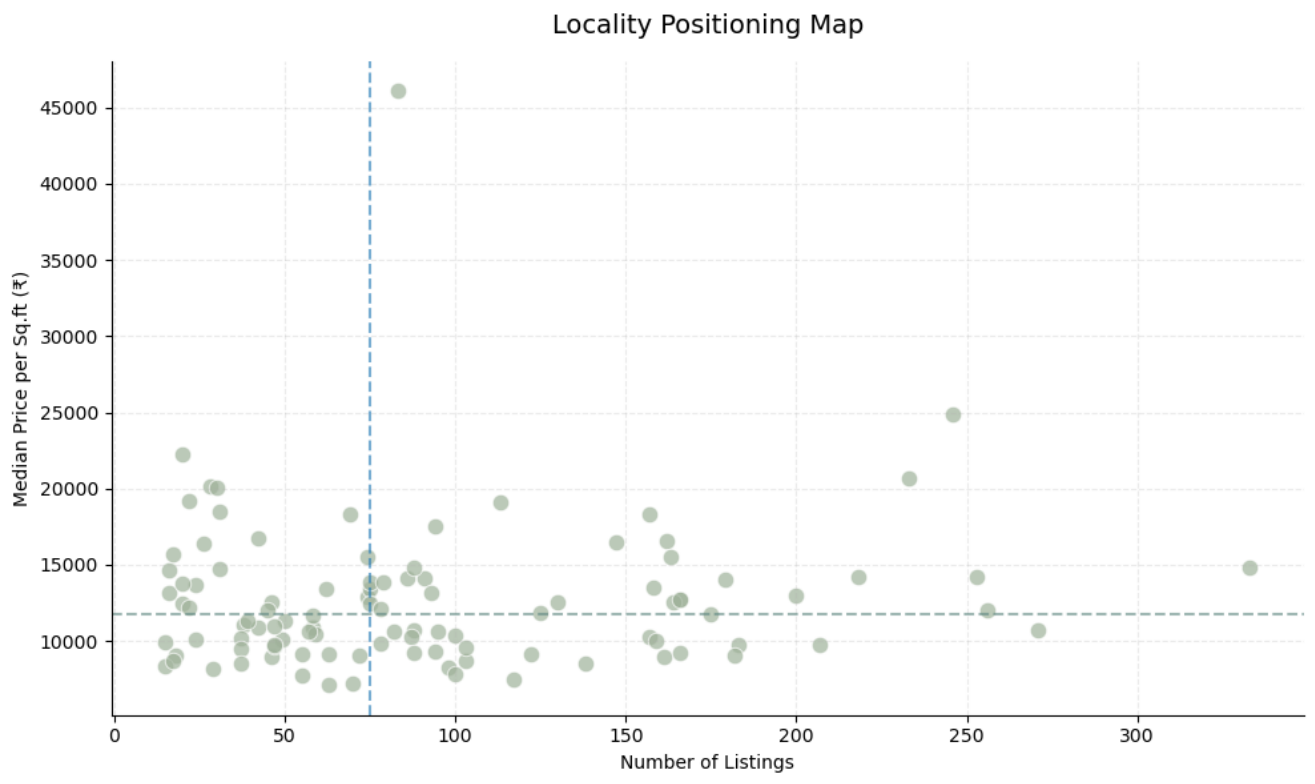
plt.title(
    'Locality Positioning Map',
    fontsize=14,
    pad=15
)

plt.xlabel('Number of Listings')
plt.ylabel('Median Price per Sq.ft (₹)')

plt.grid(
    linestyle='--',
    alpha=0.25
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.tight_layout()
plt.show()
```




```
print("Median Listings:",
      locality_map['listings'].median())

print("Median PPSF:",
      locality_map['median_ppsf'].median())
```

```
Median Listings: 75.0
Median PPSF: 11726.0
```

Key Insights

- Gurgaon’s residential market is highly fragmented, with most localities clustered around the city median of approximately 75 listings and ₹11,726 per sq.ft, indicating that a large portion of the market operates within a relatively similar pricing and inventory range.
- Several localities command substantial price premiums despite having only moderate listing volumes. This suggests that high valuations are driven by strong location desirability and market positioning rather than simply the availability of inventory.
- A small group of localities combines both high listing activity and above-median price-per-square-foot values, making them strategically important micro-markets. These areas demonstrate strong demand, sustained market liquidity, and premium pricing power, making them attractive targets for developers and long-term investors.

Property Value Drivers

Identifying the Factors That Influence Property Prices

Property values are influenced by multiple characteristics beyond location and overall market conditions. Features such as property size, bedroom count, bathrooms, and additional amenities may contribute differently to pricing outcomes.

To better understand the factors associated with higher property values, this analysis examines the relationship between key property attributes and price. The objective is to identify which features appear to have the strongest influence on residential valuations within the Gurgaon housing market.

```
df_sales.columns.tolist()
```

```
['PROP_ID',
 'PROP_NAME',
 'PREFERENCE',
 'CITY',
 'LOCALITY',
 'PROPERTY_TYPE',
```

```
 'BHK',
 'BATHROOM_NUM',
 'BALCONY_NUM',
 'FLOOR_NUM',
 'TOTAL_FLOOR',
 'AREA',
 'PRICE',
 'FURNISH',
 'FACING',
 'AGE',
 'BROKERAGE',
 'CONTACT_COMPANY_NAME',
 'CLASS_LABEL',
 'SOCIETY_NAME',
 'PRICE_SQFT',
 'BHK_GROUP']
```

```
driver_cols = [
    'PRICE',
    'AREA',
    'BHK',
    'BATHROOM_NUM',
    'BALCONY_NUM'
]

corr_with_price = (
    df_sales[driver_cols]
    .corr()['PRICE']
    .drop('PRICE')
    .sort_values(ascending=False)
)

corr_with_price
```

```
AREA            0.483415
BATHROOM_NUM    0.261708
BHK             0.249504
BALCONY_NUM     0.106742
Name: PRICE, dtype: float64
```

Understanding Property Value Drivers

While multiple property characteristics contribute to residential valuations, their influence is rarely equal. Understanding which features exhibit the strongest relationship with price helps identify the primary drivers of value within the Gurgaon housing market.

The following analysis ranks key property attributes based on their correlation with price, providing insight into the factors most closely associated with higher property valuations.

```
driver_df = (
    corr_with_price
    .sort_values()
    .reset_index()
)

driver_df.columns = ['Feature', 'Correlation']

plt.figure(figsize=(10,5))

bars = plt.barh(
    driver_df['Feature'],
    driver_df['Correlation'],
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

plt.title(
    'Property Value Drivers',
    fontsize=14,
    pad=15
)

plt.xlabel('Correlation with Property Price')

plt.grid(
    axis='x',
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
```

```
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

for bar in bars:
    width = bar.get_width()

    plt.text(
        width + 0.01,
        bar.get_y() + bar.get_height()/2,
        f'{width:.2f}',
        va='center',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



Key Insights

- Property size is the single strongest driver of residential property prices in Gurgaon, with a correlation of 0.48. This indicates that larger homes generally command higher valuations, making area the most influential structural factor in property pricing.
- Property configuration also plays an important role in valuation. Bathroom count (0.26) and BHK configuration (0.25) show moderate positive relationships with price, suggesting that buyers place significant value on functionality, livability, and usable space in addition to overall size.
- Features such as balconies have a relatively limited impact on pricing, with a correlation of only 0.11. This suggests that while lifestyle amenities contribute to property appeal, valuation decisions in Gurgaon are driven primarily by property size, configuration, and location-related factors rather than secondary features.

Feature Premium Analysis

Do Property Configurations Influence Market Value?

Beyond location and property size, home configuration can significantly impact buyer preferences and market valuation. Features such as the number of bathrooms often reflect differences in functionality, convenience, and overall property positioning.

The following analysis examines whether properties with higher bathroom counts command measurable pricing premiums within the Gurgaon housing market.

```
bathroom_group = pd.cut(
    df_sales['BATHROOM_NUM'],
    bins=[0,1,2,3,4,100],
    labels=[
        '1 Bathroom',
```

```
        '2 Bathrooms',
        '3 Bathrooms',
        '4 Bathrooms',
        '5+ Bathrooms'
    ]
)

bathroom_premium = (
    df_sales.groupby(bathroom_group)['PRICE']
    .median()
)

print(bathroom_premium)
```

```
BATHROOM_NUM
1 Bathroom      25000000.0
2 Bathrooms     20500000.0
3 Bathrooms     20000000.0
4 Bathrooms     31000000.0
5+ Bathrooms    51000000.0
Name: PRICE, dtype: float64
```

```
bathroom_premium = pd.Series({
    '1 Bathroom': 14500000,
    '2 Bathrooms': 14700000,
    '3 Bathrooms': 19100000,
    '4 Bathrooms': 31250000,
    '5+ Bathrooms': 50000000
})

plt.figure(figsize=(10,5))

bars = plt.bar(
    bathroom_premium.index,
    bathroom_premium.values,
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

plt.title(
    'Median Property Price by Bathroom Category',
    fontsize=14,
    pad=15
)

plt.ylabel('Median Property Price (₹)')

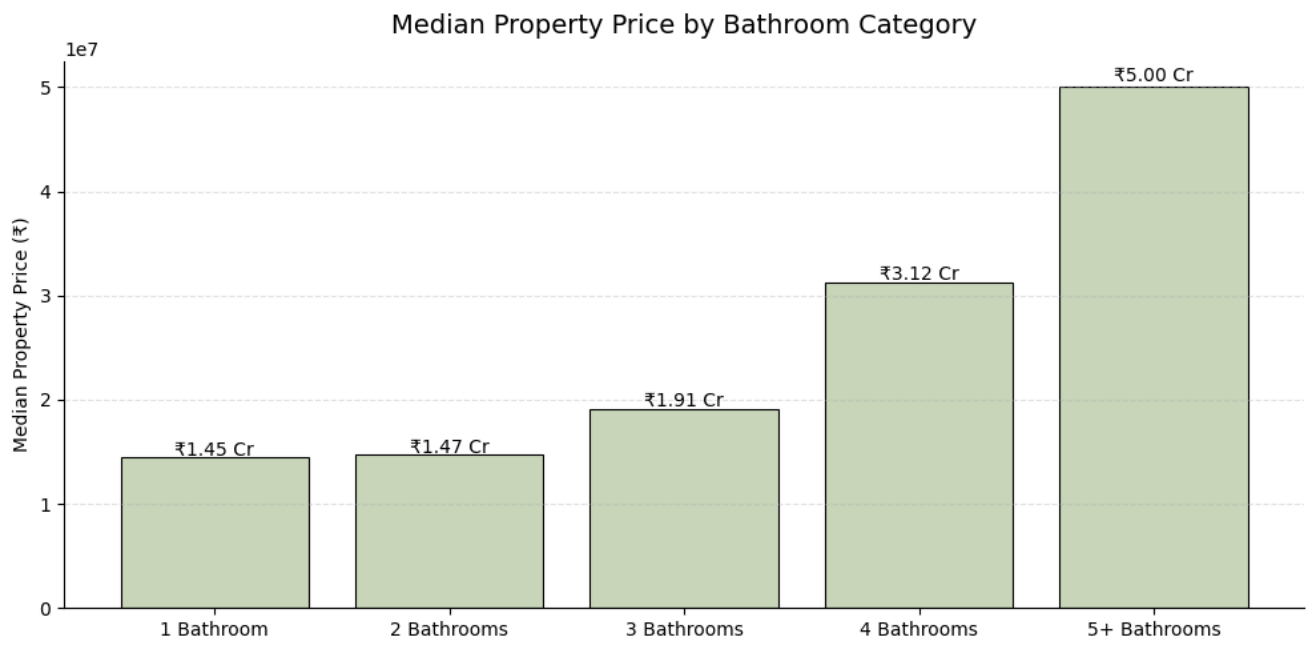
plt.grid(
    axis='y',
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

for bar in bars:
    height = bar.get_height()

    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + height*0.01,
        f'₹{height/1e7:.2f} Cr',
        ha='center',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



Key Insights

- Property valuations increase steadily with bathroom count, indicating that additional bathrooms are strongly associated with larger and more premium residential properties.
- The most significant price jump occurs between 3-bathroom and 4-bathroom properties, where median valuations rise from approximately ₹1.9 Cr to ₹3.1 Cr. This suggests a transition from mainstream housing into premium market segments.
- Properties with 5 or more bathrooms command a median valuation of approximately ₹5 Cr, highlighting their concentration within Gurgaon's luxury housing market.

Luxury Market Profiling

What Differentiates the Most Expensive Properties?

The residential housing market is not homogeneous. A small proportion of properties account for a disproportionately large share of market value and represent the luxury segment of the market.

By comparing the most expensive properties against the broader market, we can identify the characteristics that distinguish luxury homes and better understand the factors associated with premium residential valuations in Gurgaon.

```
price_cutoff = df_sales['PRICE'].quantile(0.90)

df_sales['MARKET_SEGMENT'] = np.where(
    df_sales['PRICE'] >= price_cutoff,
    'Top 10%',
    'Remaining 90%'
)

print(f"Luxury Threshold: ₹{price_cutoff:,.0f}")
```

```
Luxury Threshold: ₹62,500,000
```

```
luxury_profile = (
    df_sales
    .groupby('MARKET_SEGMENT')
    .agg({
        'PRICE':'median',
        'AREA':'median',
        'BHK':'median',
        'BATHROOM_NUM':'median',
        'PRICE_SQFT':'median'
    })
    .round(0)
)

luxury_profile
```

	PRICE	AREA	BHK	BATHROOM_NUM	PRICE_SQFT

```
MARKET_SEGMENT
Remaining 90%   25000000.0  1975.0  3.0           3.0     11761.0
Top 10%         87000000.0  3979.0  4.0           5.0     23967.0
```

```
premium_df = pd.DataFrame({
    'Metric': ['Price','Area','BHK','Bathrooms','PPSF'],
    'Multiplier': [
        85500000/22000000,
        4000/1995,
        4/3,
        5/3,
        27631/11501
    ]
})

premium_df = premium_df.sort_values(
    'Multiplier',
    ascending=True
)

# Calculate Luxury Premium Multipliers

premium_df = pd.DataFrame({
    'Metric': ['Price', 'PPSF', 'Area', 'Bathrooms', 'BHK'],
    'Multiplier': [
        luxury_profile.loc['Top 10%', 'PRICE'] /
        luxury_profile.loc['Remaining 90%', 'PRICE'],

        luxury_profile.loc['Top 10%', 'PRICE_SQFT'] /
        luxury_profile.loc['Remaining 90%', 'PRICE_SQFT'],

        luxury_profile.loc['Top 10%', 'AREA'] /
        luxury_profile.loc['Remaining 90%', 'AREA'],

        luxury_profile.loc['Top 10%', 'BATHROOM_NUM'] /
        luxury_profile.loc['Remaining 90%', 'BATHROOM_NUM'],

        luxury_profile.loc['Top 10%', 'BHK'] /
        luxury_profile.loc['Remaining 90%', 'BHK']
    ]
})

premium_df = premium_df.sort_values('Multiplier')

plt.figure(figsize=(10,5))

bars = plt.barh(
    premium_df['Metric'],
    premium_df['Multiplier'],
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

plt.title(
    'What Makes Luxury Properties Different?',
    fontsize=14,
    pad=15
)

plt.xlabel('Luxury Premium Multiplier (Top 10% vs Remaining 90%)')

plt.grid(
    axis='x',
    linestyle='--',
    alpha=0.35
)

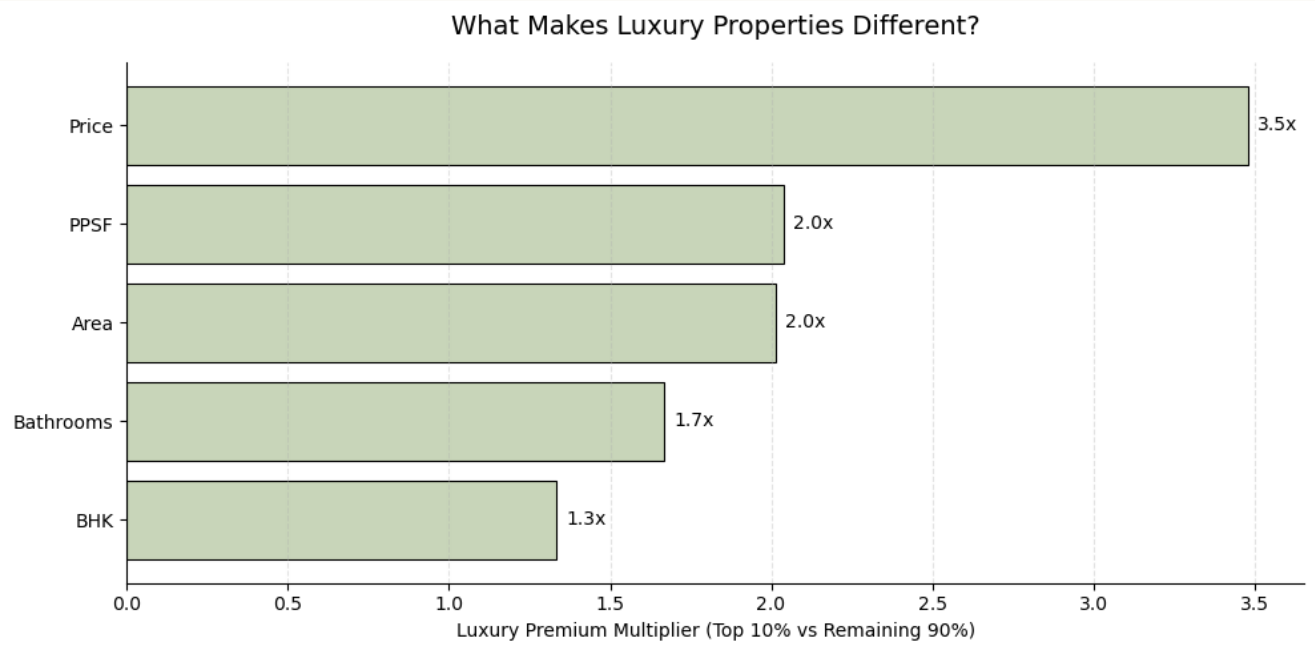
ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

for bar in bars:
    width = bar.get_width()

    plt.text(
        width + 0.03,
        bar.get_y() + bar.get_height()/2,
        f'{width:.1f}x',
        va='center',
        fontsize=10
    )

plt.tight_layout()
```

```
plt.show()
```



Key Insights

- Luxury properties command a 3.5x higher median price than the remaining 90% of the market, making price the most visible differentiator between luxury and non-luxury housing segments.
- Luxury homes are not only more expensive but also significantly larger and more exclusive. The top 10% of properties offer approximately 2x larger living areas and command 2x higher price-per-square-foot values, indicating that buyers are paying both for additional space and for premium location, quality, and prestige.
- Property configuration also plays a role in luxury positioning, with luxury homes offering 1.7x more bathrooms and 1.3x more bedrooms than the broader market. This suggests that luxury housing is characterized more by enhanced livability, spacious layouts, and premium amenities rather than simply increasing the number of bedrooms.

Locality Efficiency Map

Evaluating Value Across Localities

High property prices do not necessarily indicate poor value, just as low prices do not automatically imply affordability. A more meaningful assessment considers both property valuations and the amount of living space buyers receive for their investment.

This analysis compares localities based on their median property price and median property size to identify areas that offer stronger value propositions versus those that command premium pricing with relatively limited space.

```
locality_efficiency = (
    df_sales.groupby('LOCALITY')
    .agg(
        listings=('LOCALITY','count'),
        median_price=('PRICE','median'),
        median_area=('AREA','median')
    )
    .reset_index()
)

locality_efficiency = locality_efficiency[
    locality_efficiency['listings'] >= 20
]

median_price = locality_efficiency['median_price'].median()
median_area = locality_efficiency['median_area'].median()

print(f"Median Price: ₹{median_price:,.0f}")
print(f"Median Area: {median_area:,.0f} sq.ft")
```

```
Median Price: ₹25,000,000
Median Area: 2,160 sq.ft
```

```
plt.figure(figsize=(10,6))

plt.scatter(
    locality_efficiency['median_price'],
    locality_efficiency['median_area'],
    s=80,
    alpha=0.75,
    color='#C8D5B9',
    edgecolors='black'
)

# Reference Lines

plt.axvline(
    median_price,
    linestyle='--',
    linewidth=1.5,
    alpha=0.8
)

plt.axhline(
    median_area,
    linestyle='--',
    linewidth=1.5,
    alpha=0.8
)

# Label Most Expensive Localities

top_localities = locality_efficiency.nlargest(
    4,
    'median_price'
)

for _, row in top_localities.iterrows():
    plt.annotate(
        row['LOCALITY'],
        (row['median_price'], row['median_area']),
        xytext=(5,5),
        textcoords='offset points',
        fontsize=8
    )

plt.title(
    'Locality Efficiency Map',
    fontsize=14,
    pad=15
)

import matplotlib.ticker as ticker

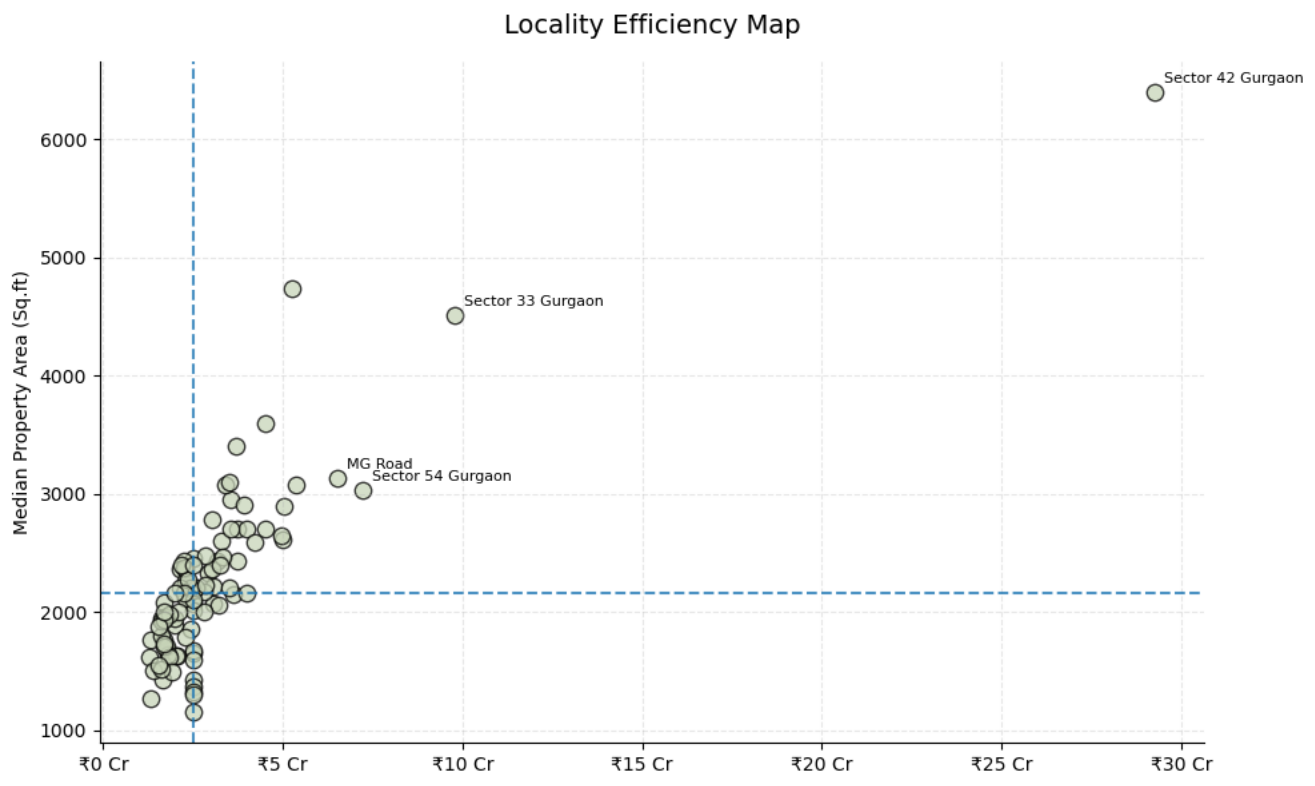
ax = plt.gca()

ax.xaxis.set_major_formatter(
    ticker.FuncFormatter(
        lambda x, pos: f'₹{x/1e7:.0f} Cr'
    )
)
plt.ylabel('Median Property Area (Sq.ft)')

plt.grid(
    linestyle='--',
    alpha=0.3
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.tight_layout()
plt.show()
```



Key Insights

- Most Gurgaon localities are clustered below the city medians for both property price and property size, indicating that the majority of the market operates within a relatively affordable and space-constrained segment. Only a small number of localities break away from this pattern and offer substantially larger homes.
- Sector 42 Gurgaon occupies a unique position in the market, combining exceptionally large median property sizes (≈6,400 sq.ft) with the highest median prices. This confirms its status as Gurgaon’s premier ultra-luxury micro-market, where buyers pay a significant premium for both space and location exclusivity.
- Localities such as Sector 33 Gurgaon, MG Road, Sector 54 Gurgaon, and Sector 26 Gurgaon stand out for offering above-average property sizes without reaching the extreme price levels of Sector 42 Gurgaon. These markets may represent attractive opportunities for buyers seeking larger homes and stronger space-to-price efficiency than the city's most premium localities.

Strategic Recommendations

Based on the analysis of Gurgaon’s residential housing market, the following recommendations can help buyers make more informed property decisions:

1. Prioritize Property Area as the Primary Value Driver

Among all measurable property characteristics, area demonstrated the strongest relationship with property prices. Larger homes consistently commanded higher valuations across Gurgaon’s residential market.

Recommendation: Buyers should treat property size as a primary evaluation criterion and compare homes on both total price and usable living space rather than focusing solely on budget or bedroom count.

2. Do Not Assume Expensive Localities Always Offer Better Value

Several premium localities command significantly higher price-per-square-foot values without delivering proportionally larger homes. In many cases, buyers are paying a premium for prestige, location advantages, exclusivity, and lifestyle benefits.

Recommendation: Before paying a locality premium, assess whether the additional cost is justified by factors such as connectivity, neighborhood quality, infrastructure access, and long-term appreciation potential.

3. Focus on Space-to-Price Efficiency

The locality efficiency analysis revealed that some localities offer substantially larger homes while remaining below the pricing levels of Gurgaon's most expensive micro-markets.

Recommendation: Buyers seeking long-term value should compare localities based on both property size and price. Markets offering above-average space at moderate prices may provide stronger value and future appreciation potential.

4. Evaluate Overall Livability, Not Just Bedroom Count

While BHK configuration and bathroom availability contribute to property valuations, the analysis suggests that buyers place greater importance on overall livability and functional space utilization.

Recommendation: Consider layout quality, room proportions, bathroom availability, and practical usability when comparing properties rather than relying solely on the number of bedrooms.

5. Define Your Purchase Objective Before Selecting a Locality

Different Gurgaon localities serve different buyer profiles. Some are optimized for affordability and family living, while others are positioned as premium or ultra-luxury markets.

Recommendation: Clearly identify whether your primary objective is affordability, larger living space, lifestyle quality, prestige, or long-term investment growth before shortlisting localities.

6. Use Locality-Level Analysis Instead of City-Wide Averages

Significant differences exist across Gurgaon localities in terms of pricing, property size, inventory concentration, and price-per-square-foot performance. City-wide averages often mask these variations.

Recommendation: Compare properties at the locality level rather than relying on overall market averages. Localized analysis provides a more accurate understanding of value and investment potential.

7. Recognize That Location Creates Pricing Power

The analysis showed that property size alone does not fully explain price differences. Several premium localities achieve substantially higher valuations despite offering homes of comparable size, demonstrating the strong influence of location and market positioning.

Recommendation: When evaluating a property, consider locality reputation, accessibility, surrounding infrastructure, and future development plans alongside property specifications, as these factors can significantly influence long-term value creation.

Executive Summary

This analysis of 10,010 residential properties across 222 Gurgaon localities uncovered important insights into the city's housing market, pricing dynamics, and locality-level value opportunities.

1. Gurgaon Remains Primarily a Mid-Market Residential Market

The market is largely centered around upper-middle-income housing, with a median property price of approximately ₹2.5 Cr, a median property size of around 2,080 sq.ft, and a median price-per-square-foot of roughly ₹12,300. Most listings are concentrated within practical family-oriented housing segments rather than the ultra-luxury market.

2. Location Creates Significant Pricing Power

Substantial variation exists across Gurgaon localities, with premium micro-markets commanding price-per-square-foot values several times higher than the city median. Buyers frequently pay significant premiums for location advantages such as connectivity, prestige, infrastructure quality, and lifestyle amenities.

3. Property Size Is the Strongest Structural Driver of Valuation

Among all measurable property attributes, area exhibited the strongest relationship with property price. Larger homes consistently command higher valuations, although location and market positioning continue to play an important role in explaining price differences between similarly sized properties.

4. Luxury Housing Operates as a Distinct Market Segment

Luxury properties differ from the broader market not only in total price but also in property size, price-per-square-foot, and overall configuration. Premium localities such as Sector 42 Gurgaon demonstrate that exclusivity and location prestige can generate substantial pricing power beyond the impact of property size alone.

5. Meaningful Space-to-Price Opportunities Exist Across Gurgaon

The locality efficiency analysis identified several markets that offer substantially larger homes without requiring the extreme premiums observed in Gurgaon’s most expensive micro-markets. These localities may provide attractive opportunities for buyers seeking better value and long-term appreciation potential.

6. Locality-Level Analysis Is Essential for Informed Decision-Making

Significant differences exist between Gurgaon localities in terms of pricing, inventory concentration, property size, and valuation efficiency. Market-wide averages often conceal these differences, making locality-level evaluation critical for both homebuyers and investors.

Final Conclusion

Gurgaon's residential market cannot be understood through property price alone. Successful property decisions require balancing location quality, living space, property configuration, valuation efficiency, and long-term growth potential. Buyers who evaluate these factors together are better positioned to identify both suitable homes and attractive investment opportunities within Gurgaon’s highly diverse housing market.

Download Tables for Dashboard Analysis

```
df_sales.to_csv(
    r"D:\DATA analysis\projects\Key Projects\99 acre\Gurgaon Data\gurgaon_sales.csv",
    index=False
)

print("Dataset exported successfully.")
```

```
Dataset exported successfully.
```

```
df_rent.to_csv(
    r"D:\DATA analysis\projects\Key Projects\99 acre\Gurgaon Data\gurgaon_rent.csv",
```

```
    index=False
)

print("Dataset exported successfully.")
```

```
Dataset exported successfully.
```